

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]
```

```

cols = list(dict.fromkeys(target_col + features_X +
                          feats_to_balance + feats_grouping +
                          feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

return df

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r") as f:
            #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")

    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)

```

```

#Removing of rabbit 4
df["column"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features

```

```

def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")

```

Function print :

1.2 Metadata

Nombre de lignes df

8068

Nombre de lignes df clean	8068
Nombre de features	29
% de NaN	0.0
Taille dataset entraînement	5647
Taille dataset test	2420
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_3h_mean, r_NH3_st_3h_std, r_NH3_st_3h_rms, r_NH3_st_3h_median, r_NH3_st_3h_min, r_NH3_st_3h_max, r_NH3_st_3h_diff_last, r_NH3_st_3h_slope, r_temperature_st_3h_mean, r_temperature_st_3h_std, r_temperature_st_3h_rms, r_temperature_st_3h_median, r_temperature_st_3h_min, r_temperature_st_3h_max, r_temperature_st_3h_diff_last, r_temperature_st_3h_slope, r_umidity_st_3h_mean, r_umidity_st_3h_std, r_umidity_st_3h_rms, r_umidity_st_3h_median, r_umidity_st_3h_min, r_umidity_st_3h_max, r_umidity_st_3h_diff_last, r_umidity_st_3h_slope

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

The following section has been done with a threshold choosen of : 0.8

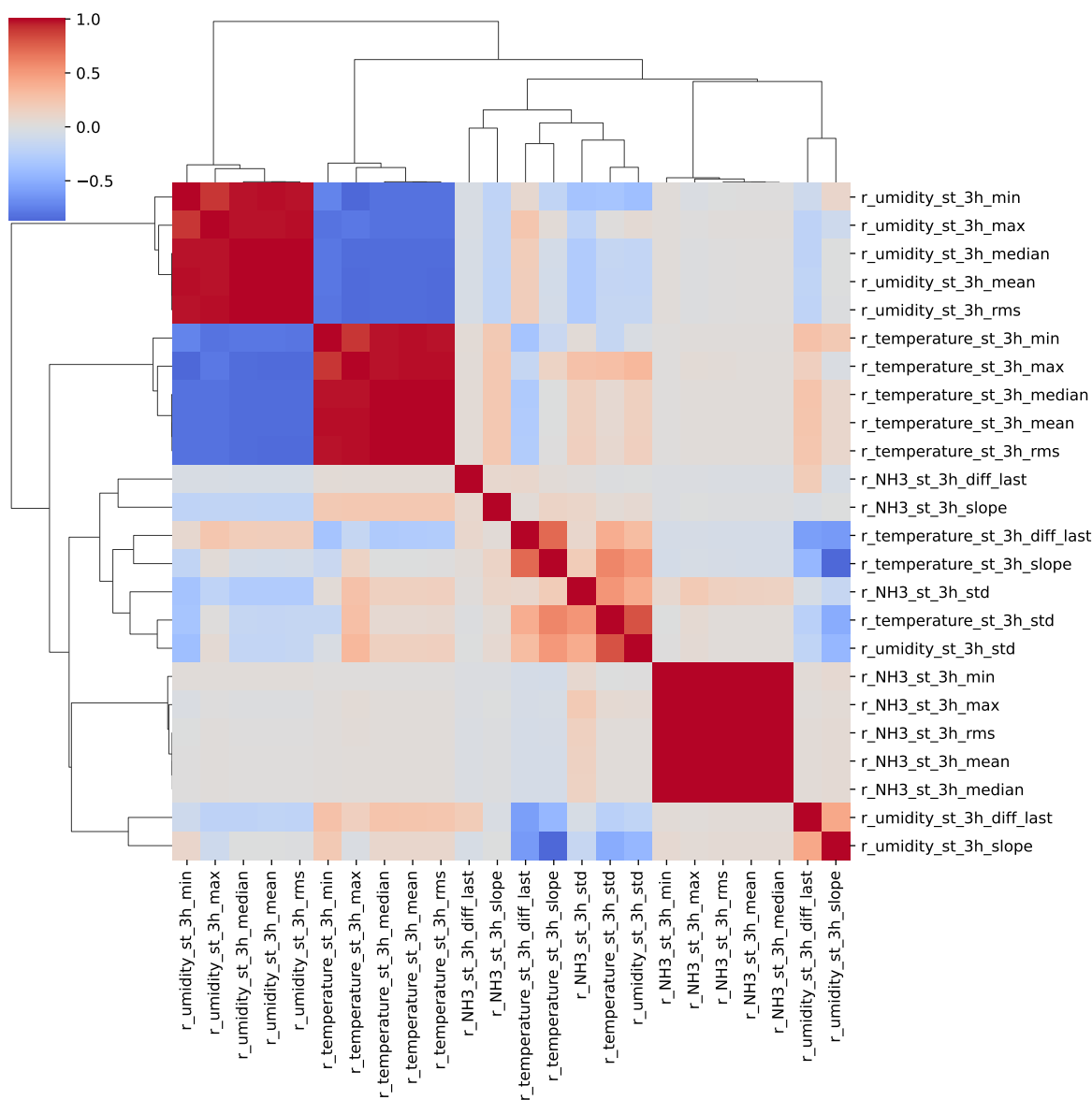
cluster 1: r, , N, H, 3 ; **cluster 2:** r, , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, , u, m, i, d, i, t, y ; **cluster 4:** r, , C, O, 2 ; **cluster 5:** r, _, T, H, I ; **cluster 6:** r_NH3_st_3h_median, r_NH3_st_3h_max, r_NH3_st_3h_mean, r_NH3_st_3h_rms, r_NH3_st_3h_min ; **cluster 7:** r_NH3_st_3h_std ; **cluster 8:** r_NH3_st_3h_diff_last ; **cluster 9:** r_NH3_st_3h_slope ; **cluster 10:** r_temperature_st_3h_mean, r_umidity_st_3h_max, r_umidity_st_3h_mean, r_umidity_st_3h_rms, r_temperature_st_3h_median, r_umidity_st_3h_min, r_umidity_st_3h_median, r_temperature_st_3h_min, r_temperature_st_3h_max, r_temperature_st_3h_rms ; **cluster 11:** r_umidity_st_3h_std, r_temperature_st_3h_std ; **cluster 12:** r_temperature_st_3h_diff_last ; **cluster**

13: r_temperature_st_3h_slope, r_umidity_st_3h_slope ; **cluster 14:** r_umidity_st_3h_diff_last ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Choosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_3h_median, r_NH3_st_3h_std, r_NH3_st_3h_diff_last, r_NH3_st_3h_slope, r_temperature_st_3h_mean, r_umidity_st_3h_std, r_temperature_st_3h_diff_last, r_temperature_st_3h_slope, r_umidity_st_3h_diff_last



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_3h_median, r_NH3_st_3h_std, r_NH3_st_3h_diff_last, r_NH3_st_3h_slope, r_tempera-

ture_st_3h_mean, r_umidity_st_3h_std, r_temperature_st_3h_diff_last, r_temperature_st_3h_slope, r_umidity_st_3h_diff_last

Len(Features__X): 14

Config Gridsearch:{‘show’: True, ‘search_type’: ‘optuna’, ‘model_name’: None, ‘model’: None, ‘param_grid’: None, ‘best_param’: None, ‘cv_func’: ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), ‘save_models’: True, ‘sumup-table’: True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

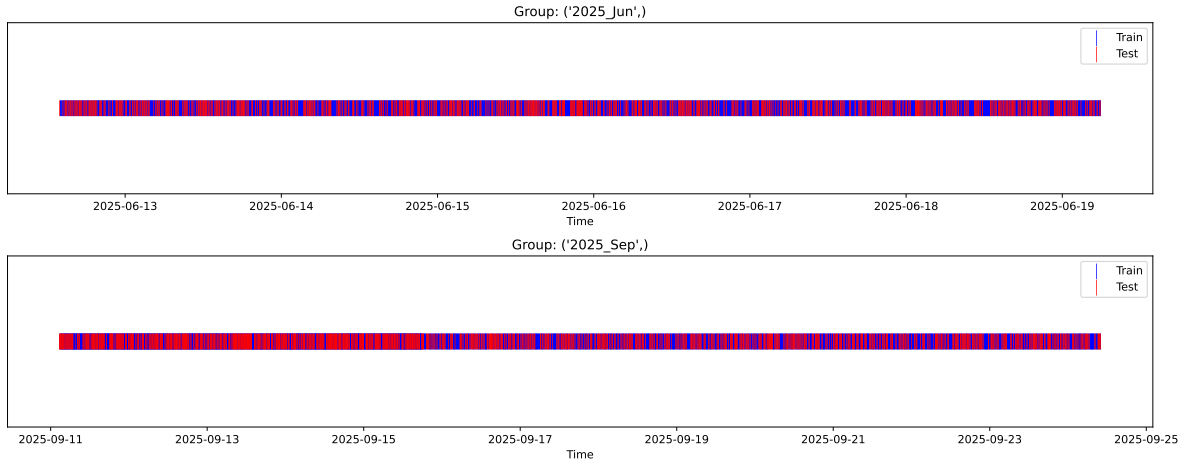
[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_train vs X_test

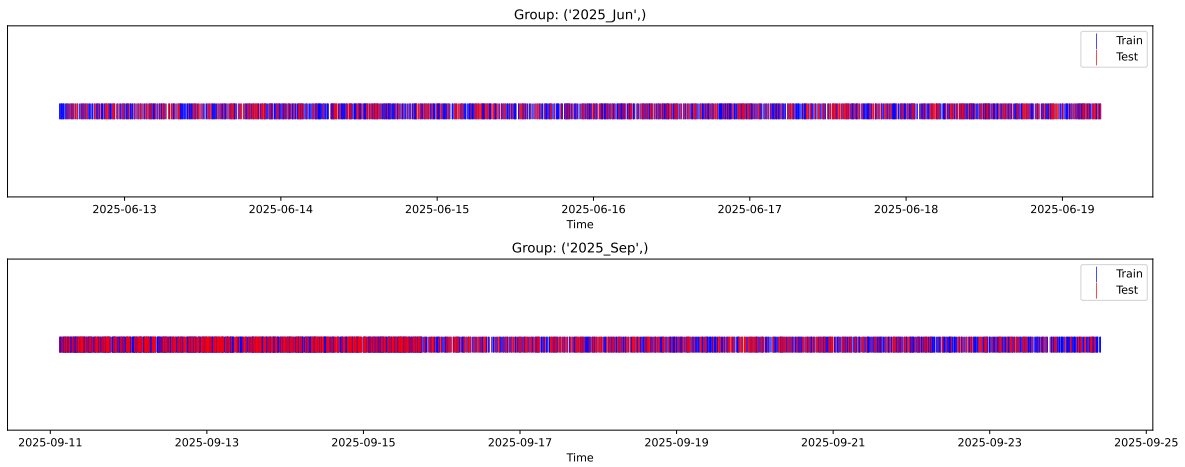
External Split: Train: 5647 samples Test: 2421 samples



3.3.2 Internal CV Folds

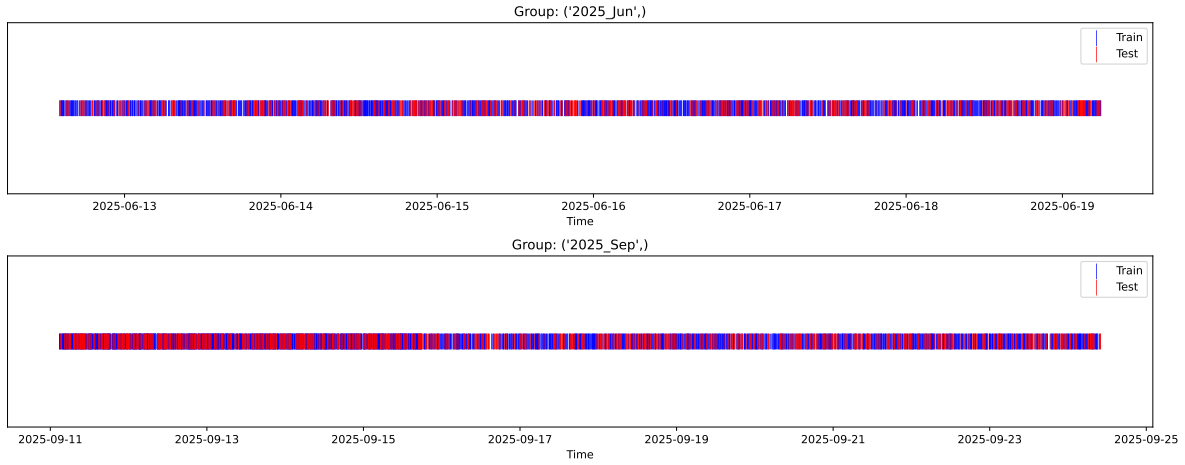
Fold 1/5

Train: 3952 samples Test: 1695 samples Excluded: 0 samples Window exclusion: 0h



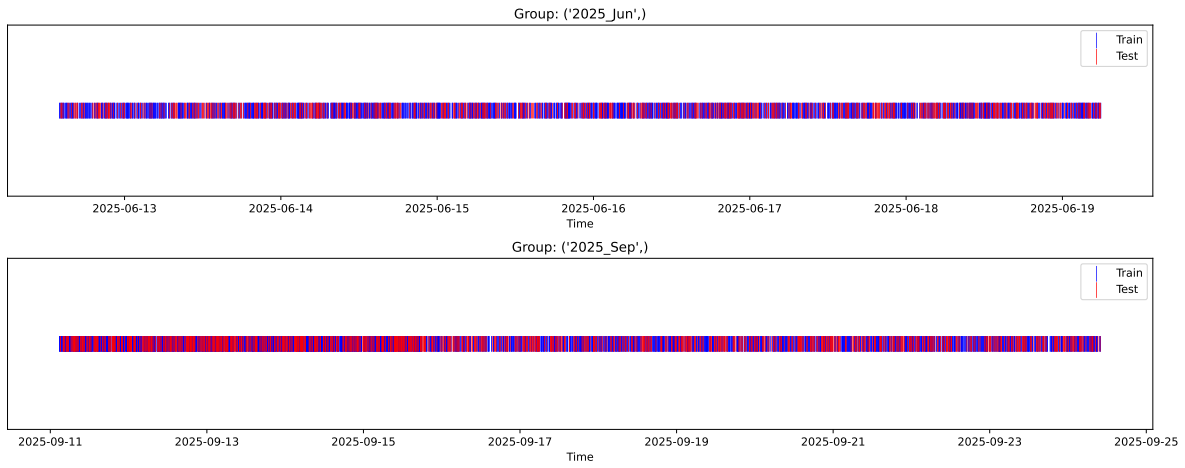
Fold 2/5

Train: 3952 samples Test: 1695 samples Excluded: 0 samples Window exclusion: 0h



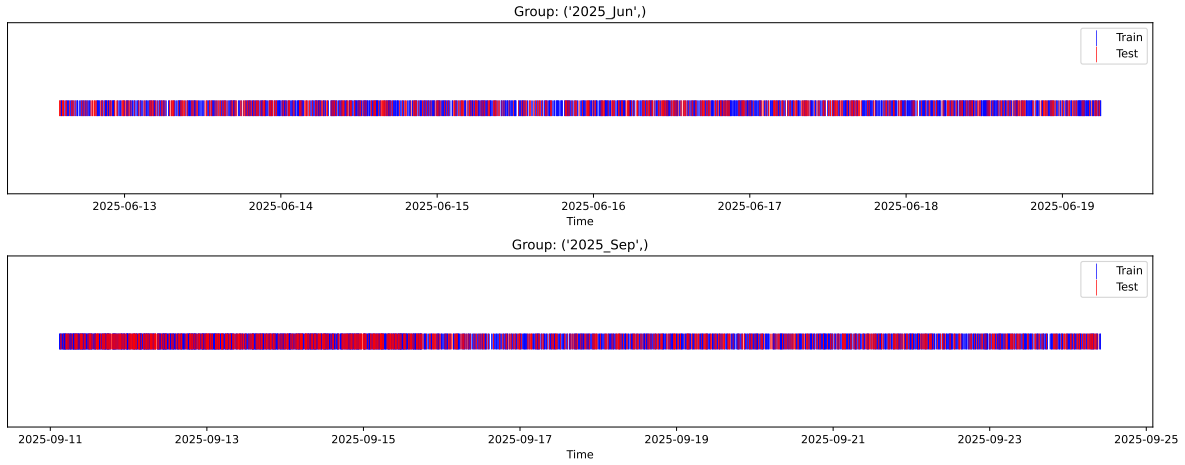
Fold 3/5

Train: 3952 samples Test: 1695 samples Excluded: 0 samples Window exclusion: 0h



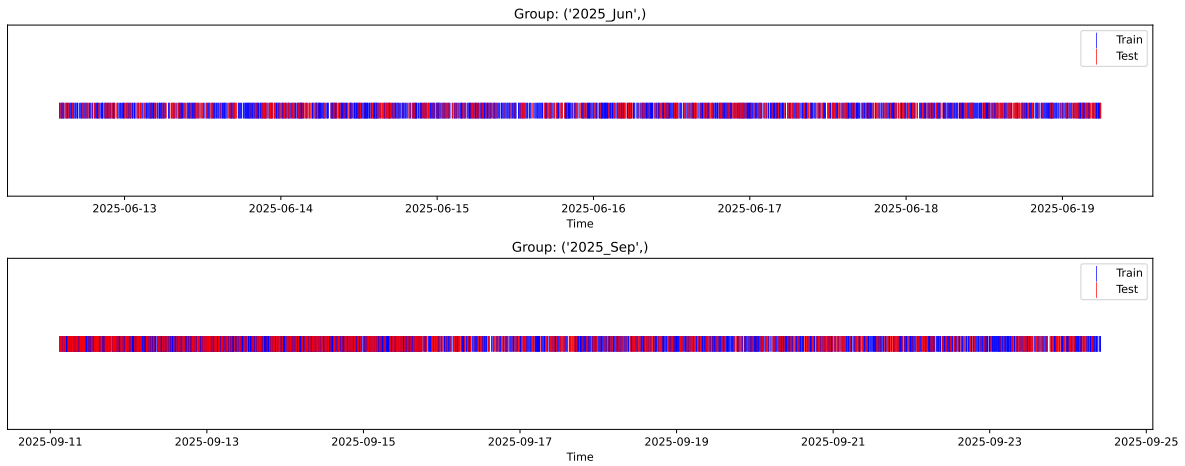
Fold 4/5

Train: 3952 samples Test: 1695 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3952 samples Test: 1695 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.347, std=0.801, min=1.041, max=8.682
 y_{test} : mean=2.362, std=0.789, min=0.959, max=13.845

Fold 0 — y_{test} : mean=2.34, std=0.78, y_{train} : mean=2.35, std=0.81

Fold 1 — y_{test} : mean=2.34, std=0.77, y_{train} : mean=2.35, std=0.81

Fold 2 — y_{test} : mean=2.35, std=0.82, y_{train} : mean=2.34, std=0.79

Fold 3 — y_test: mean=2.35, std=0.79, y_train: mean=2.34, std=0.80

Fold 4 — y_test: mean=2.34, std=0.80, y_train: mean=2.35, std=0.80

train scores CV: [0.55828295 0.55971907 0.55441816 0.55591822 0.5513032]

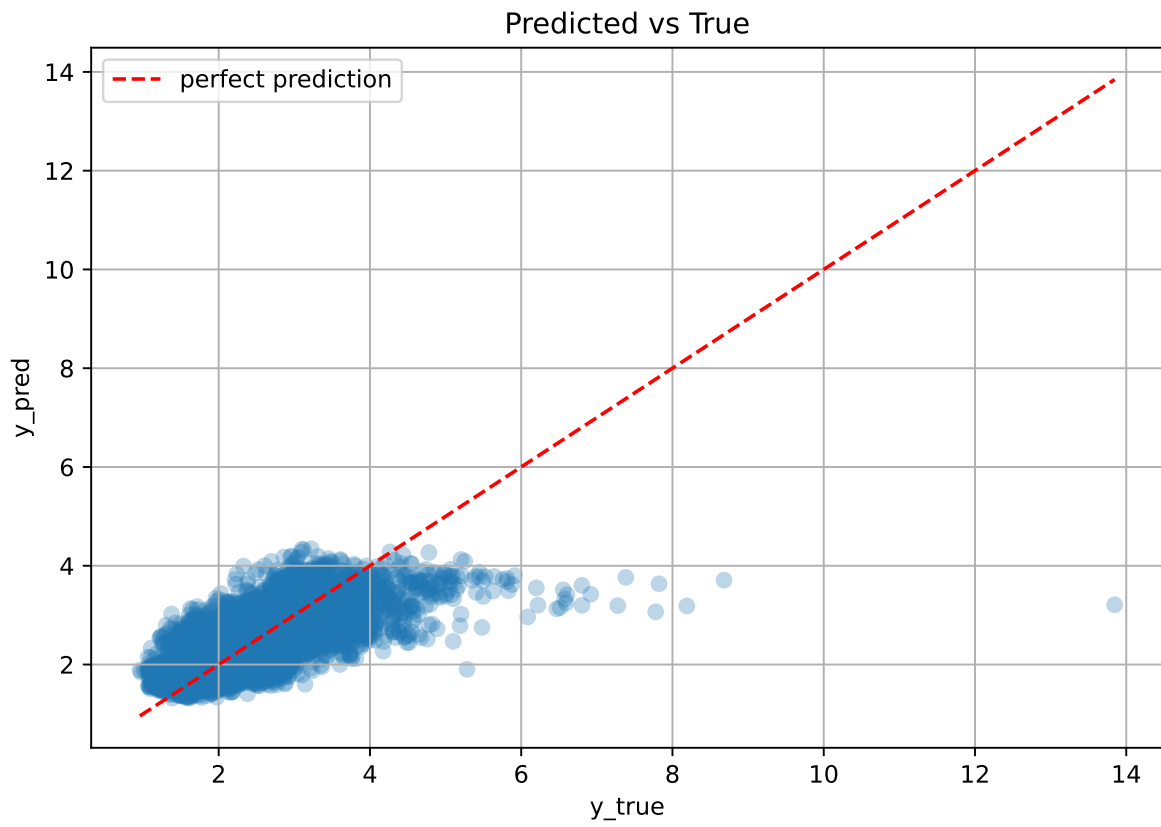
test scores CV: [0.54904644 0.54380809 0.55814877 0.55430327 0.56588647]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5559	0.5542	0.0076	0.5056	0.5067	-0.0476	0.9142	1.003

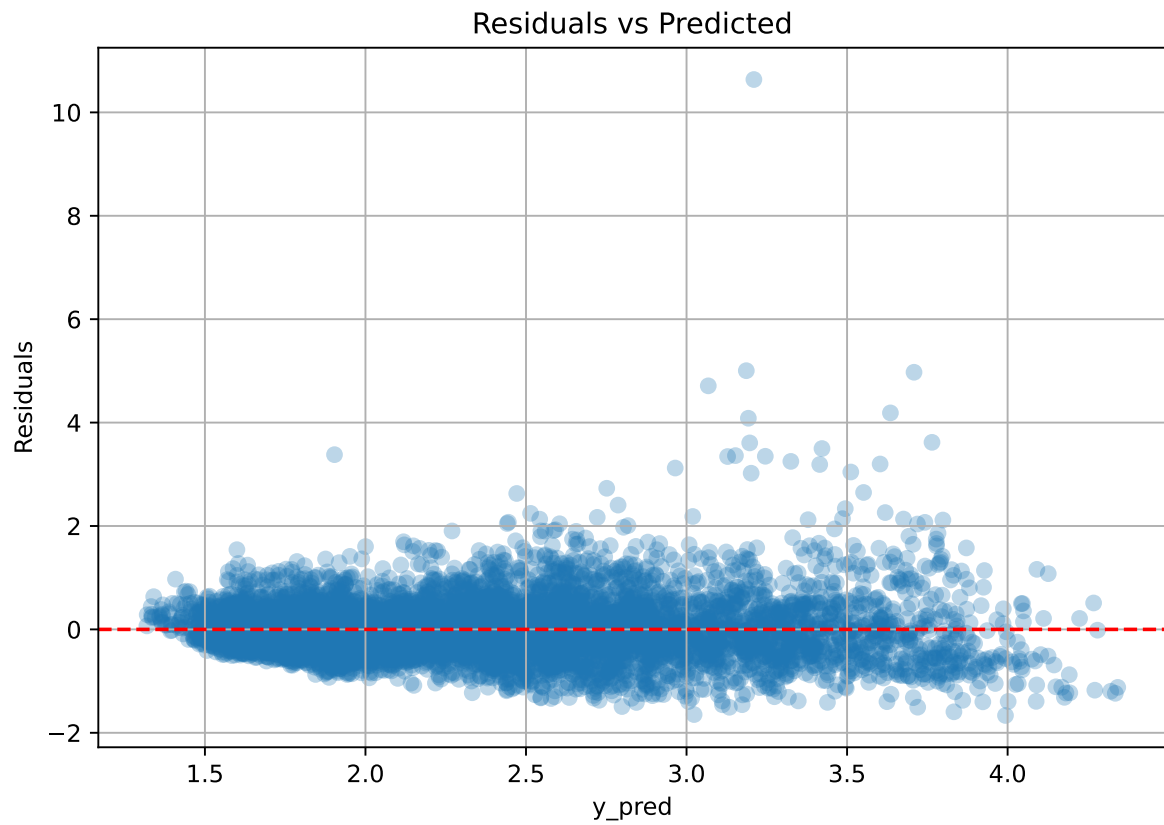
Coefficients:

coef_r_CO2 = -0.0543, coef_r_NH3 = -0.1756, coef_r_NH3_st_3h_diff_last = 0.0239, coef_r_NH3_st_3h_median = 0.1260, coef_r_NH3_st_3h_slope = 0.0282, coef_r_NH3_st_3h_std = 0.0931, coef_r_THI = 0.5894, coef_r_temperature = 0.4094, coef_r_temperature_st_3h_diff_last = -0.0630, coef_r_temperature_st_3h_mean = -1.0808, coef_r_temperature_st_3h_slope = -0.1372, coef_r_umidity = -0.5043, coef_r_umidity_st_3h_diff_last = -0.0393, coef_r_umidity_st_3h_std = 0.0435, intercept = 2.3465,

3.4.2 Scatter



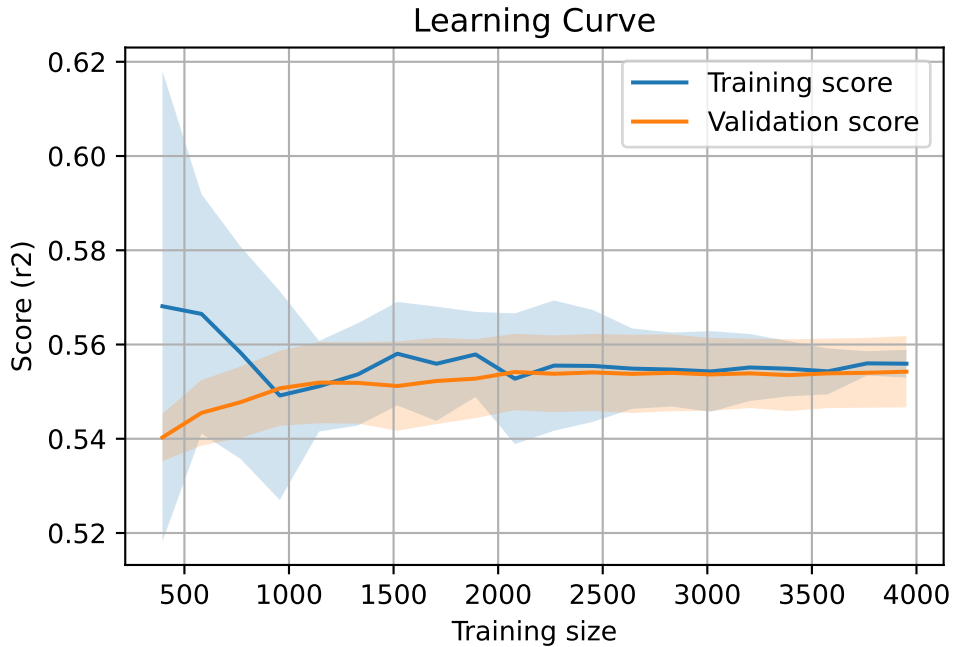
3.4.3 Scatter residuals



3.4.4 Learning curve

`Len(group)` : 8068, `Len(training)` : 5648,

`Len(training_real)` : 3954, `Len(test_of_training)` : 1694, `Len(test)` : 2420

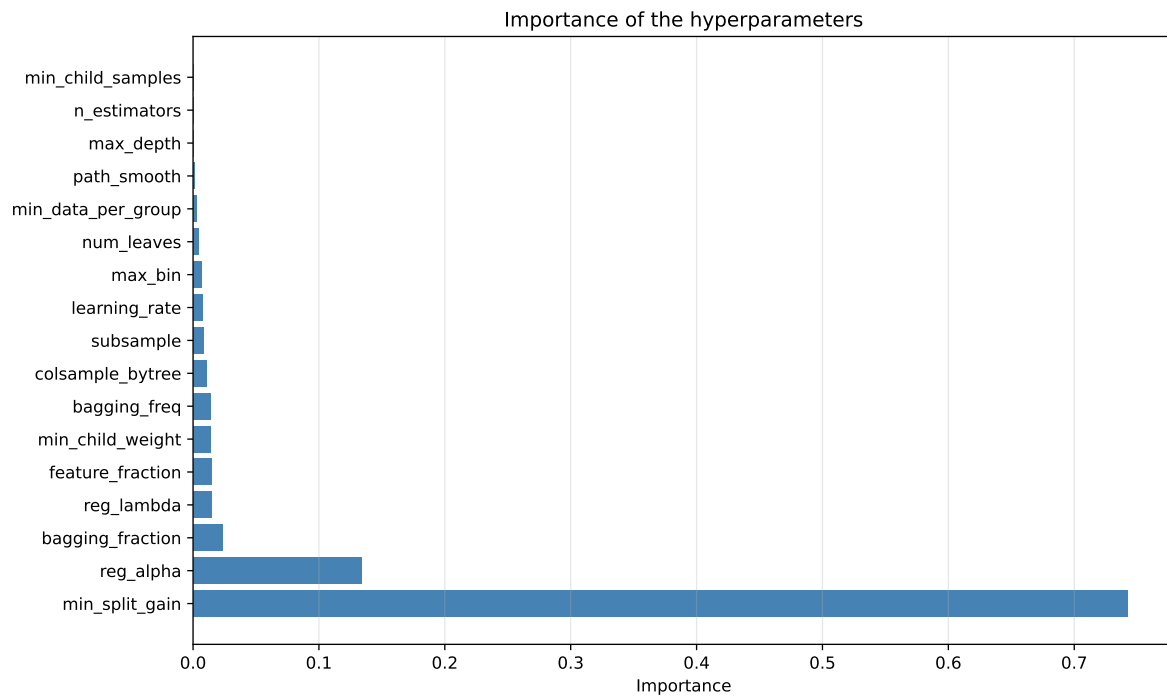
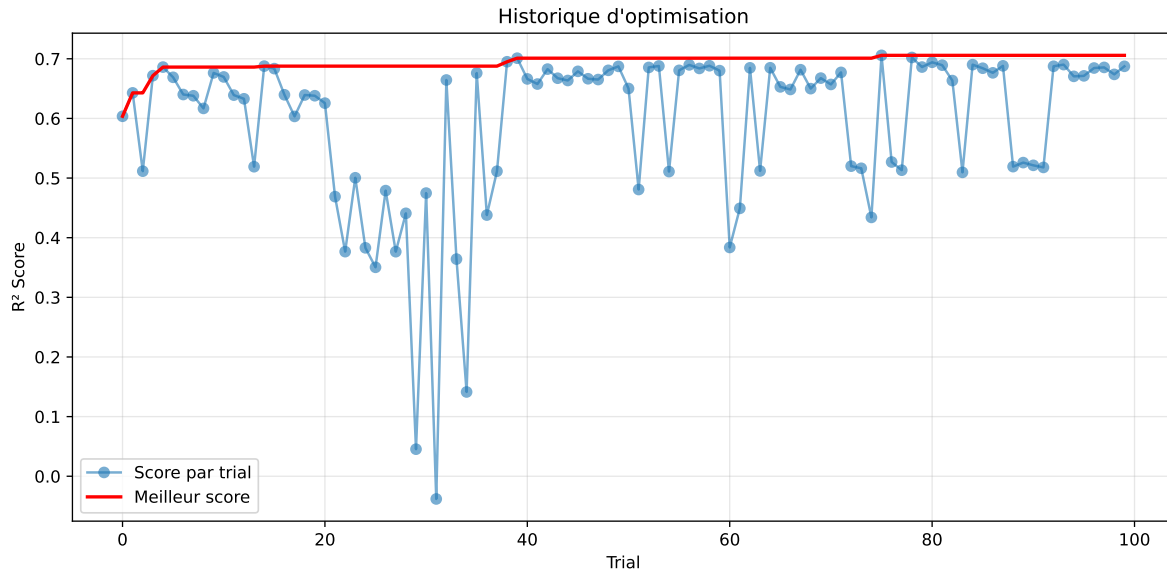


3.5 Model : LGBM

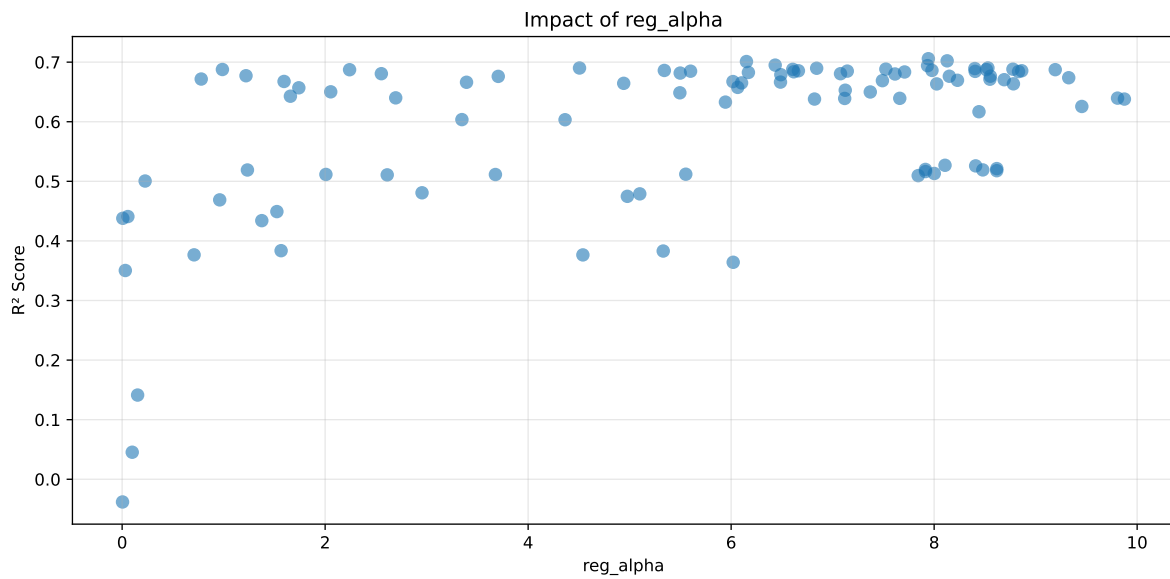
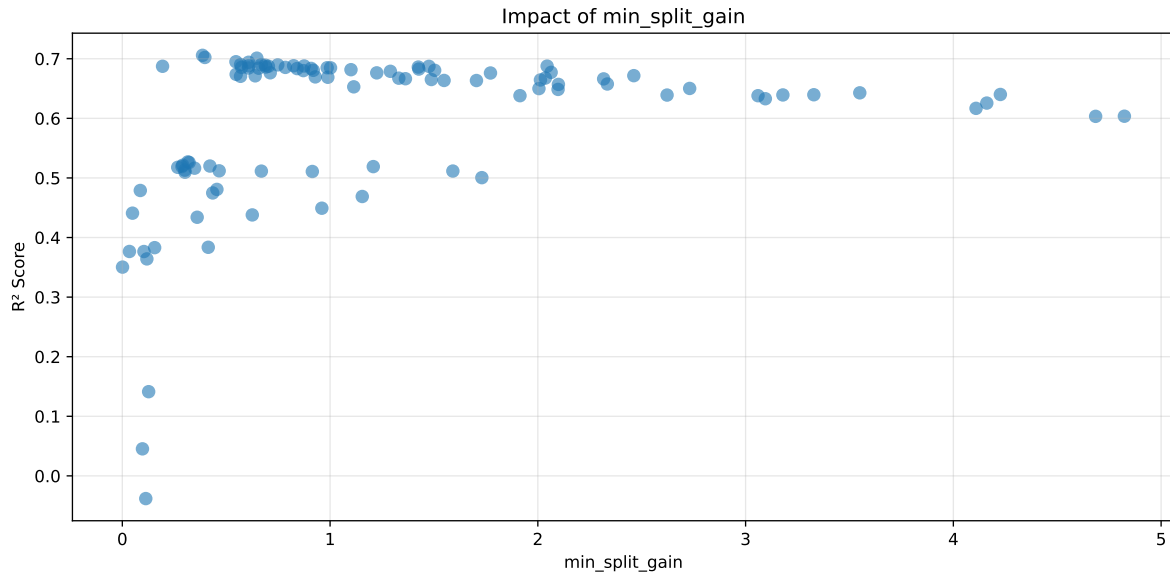
3.5.1 Gridsearch

Distribution y_train vs y_test: y_train: mean=2.347, std=0.801, min=1.041, max=8.682
y_test: mean=2.362, std=0.789, min=0.959, max=13.845

===== TOP Importance FEATURES =====
feature importance r_CO2 1200 r_NH3_st_3h_median 784 r_umidity 770 r_NH3 726
r_temperature_st_3h_slope 582 r_NH3_st_3h_slope 416 r_NH3_st_3h_std 371
r_umidity_st_3h_std 342 r_temperature 321 r_temperature_st_3h_diff_last 292 r_temper-
ature_st_3h_mean 289 r_NH3_st_3h_diff_last 260 r_THI 215 r_umidity_st_3h_diff_last
188

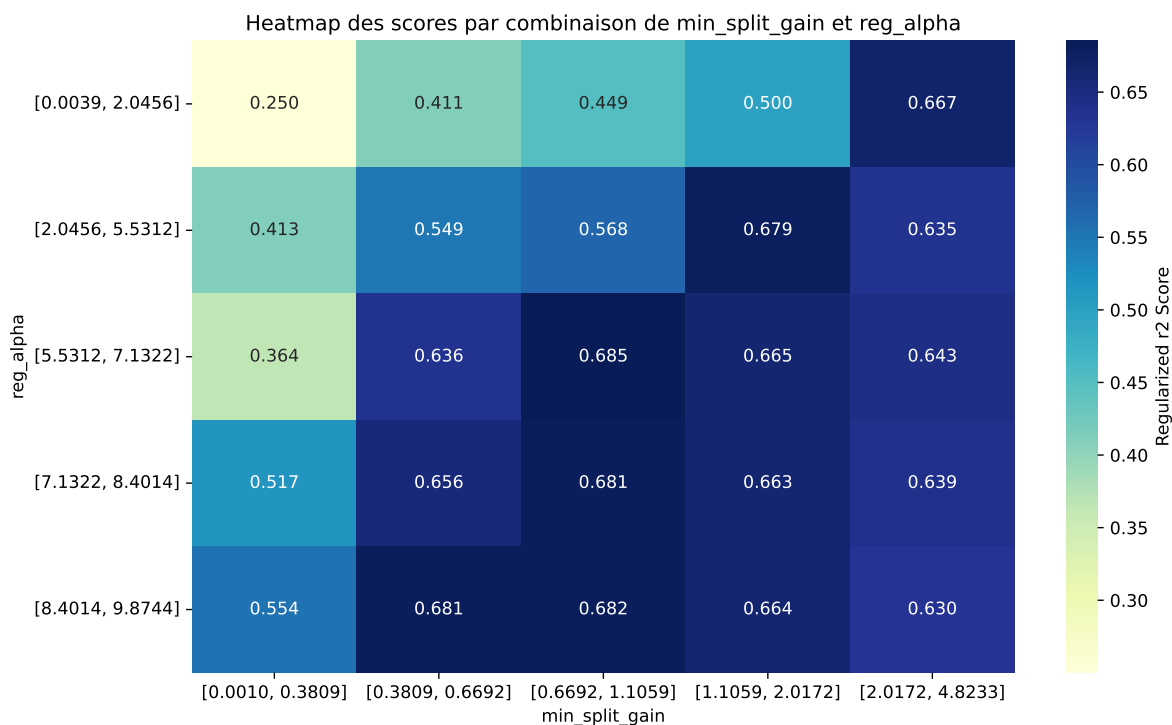


Paramètres avec >5% d'importance : ['min_split_gain', 'reg_alpha']



C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7406	0.7058	0.6624	0.0093	0.7058	0.6804	-0.0253	0.9641	1.0494
2	0.7365	0.7022	0.6597	0.0079	0.7022	0.6784	-0.0238	0.9661	1.0489
3	0.7327	0.701	0.6553	0.0101	0.701	0.6733	-0.0277	0.9605	1.0452
4	0.7252	0.695	0.6511	0.0071	0.695	0.6698	-0.0252	0.9638	1.0435
5	0.7255	0.6941	0.651	0.0084	0.6941	0.6682	-0.0259	0.9627	1.0452
6	0.7195	0.6902	0.6475	0.0077	0.6902	0.6669	-0.0234	0.9661	1.0424
7	0.7216	0.6902	0.6458	0.0099	0.6902	0.6536	-0.0366	0.947	1.0456
8	0.7196	0.6898	0.6462	0.0098	0.6898	0.665	-0.0248	0.9641	1.0432
9	0.7188	0.6893	0.6454	0.0084	0.6893	0.663	-0.0262	0.9619	1.0428
10	0.7172	0.6883	0.6442	0.0082	0.6883	0.6622	-0.0261	0.962	1.042
---	---	---	---	---	---	---	---	---	---
100	0.9298	0.7685	0.7203	0.0103	-0.0381	0.742	-0.0265	0.9655	1.2099

Hyperparameters:

Rank 1: {'reg__n_estimators': 700, 'reg__max_depth': 6, 'reg__learning_rate': 0.01609033620000133, 'reg__num_leaves': 55, 'reg__subsample': 0.7701835749647757, 'reg__colsample_bytree': 0.7898365239630291, 'reg__min_child_samples': 41, 'reg__reg_alpha': 7.943110809460675, 'reg__reg_lambda': 23.711433041694193, 'reg__min_split_gain': 0.38609023553190236, 'reg__path_smooth': 2.4641613900865758, 'reg__min_child_weight':

0.025163368234505802, 'reg__feature_fraction': 0.6372211734200639, 'reg__bagging_fraction': 0.7165061424169844, 'reg__bagging_freq': 1, 'reg__max_bin': 95, 'reg__min_data_per_group': 188}

Rank 2: {'reg__n_estimators': 700, 'reg__max_depth': 6, 'reg__learning_rate': 0.01835437169832292, 'reg__num_leaves': 45, 'reg__subsample': 0.8696886751936379, 'reg__colsample_bytree': 0.6908864347121224, 'reg__min_child_samples': 41, 'reg__reg_alpha': 8.128272410408261, 'reg__reg_lambda': 23.67328304367148, 'reg__min_split_gain': 0.39736190512348596, 'reg__path_smooth': 0.8376467028708896, 'reg__min_child_weight': 0.13243098795432257, 'reg__feature_fraction': 0.6372046492456072, 'reg__bagging_fraction': 0.6434678015905935, 'reg__bagging_freq': 2, 'reg__max_bin': 197, 'reg__min_data_per_group': 191}

Rank 3: {'reg__n_estimators': 600, 'reg__max_depth': 6, 'reg__learning_rate': 0.03126607800556019, 'reg__num_leaves': 43, 'reg__subsample': 0.7316697582164774, 'reg__colsample_bytree': 0.6165116676394709, 'reg__min_child_samples': 38, 'reg__reg_alpha': 6.1509260943147765, 'reg__reg_lambda': 21.644405461985922, 'reg__min_split_gain': 0.6475727118377782, 'reg__path_smooth': 1.2566660545852808, 'reg__min_child_weight': 0.15592834003429745, 'reg__feature_fraction': 0.6495860042493365, 'reg__bagging_fraction': 0.6552583969473793, 'reg__bagging_freq': 2, 'reg__max_bin': 223, 'reg__min_data_per_group': 131}

Rank 4: {'reg__n_estimators': 600, 'reg__max_depth': 6, 'reg__learning_rate': 0.03276262297318885, 'reg__num_leaves': 44, 'reg__subsample': 0.7396012071772914, 'reg__colsample_bytree': 0.61675254926272, 'reg__min_child_samples': 38, 'reg__reg_alpha': 6.4340234039415405, 'reg__reg_lambda': 20.50451189030842, 'reg__min_split_gain': 0.5468670038088932, 'reg__path_smooth': 1.3708192763720515, 'reg__min_child_weight': 0.13732226372746753, 'reg__feature_fraction': 0.5025651418061667, 'reg__bagging_fraction': 0.5863036004665205, 'reg__bagging_freq': 2, 'reg__max_bin': 142, 'reg__min_data_per_group': 169}

Rank 5: {'reg__n_estimators': 700, 'reg__max_depth': 5, 'reg__learning_rate': 0.015974148461465797, 'reg__num_leaves': 54, 'reg__subsample': 0.8614829968038942, 'reg__colsample_bytree': 0.5933053659739966, 'reg__min_child_samples': 50, 'reg__reg_alpha': 7.935295229816952, 'reg__reg_lambda': 13.017860079508116, 'reg__min_split_gain': 0.607636655694524, 'reg__path_smooth': 3.277430331974502, 'reg__min_child_weight': 0.13307745958123204, 'reg__feature_fraction': 0.631843285024066, 'reg__bagging_fraction': 0.7203917274920358, 'reg__bagging_freq': 2, 'reg__max_bin': 196, 'reg__min_data_per_group': 190}

Rank 6: {'reg__n_estimators': 600, 'reg__max_depth': 5, 'reg__learning_rate': 0.015931070205097885, 'reg__num_leaves': 56, 'reg__subsample': 0.8628911613197885, 'reg__colsample_bytree': 0.7831001064354945, 'reg__min_child_samples': 50, 'reg__reg_alpha': 8.529919542376142, 'reg__reg_lambda': 26.38934673495237, 'reg__min_split_gain': 0.5693859227540474, 'reg__path_smooth': 3.4165164039225755, 'reg__min_child_weight':

0.1269645173356267, 'reg__feature_fraction': 0.626257158368274, 'reg__bagging_fraction': 0.712656971017997, 'reg__bagging_freq': 2, 'reg__max_bin': 216, 'reg__min_data_per_group': 182}

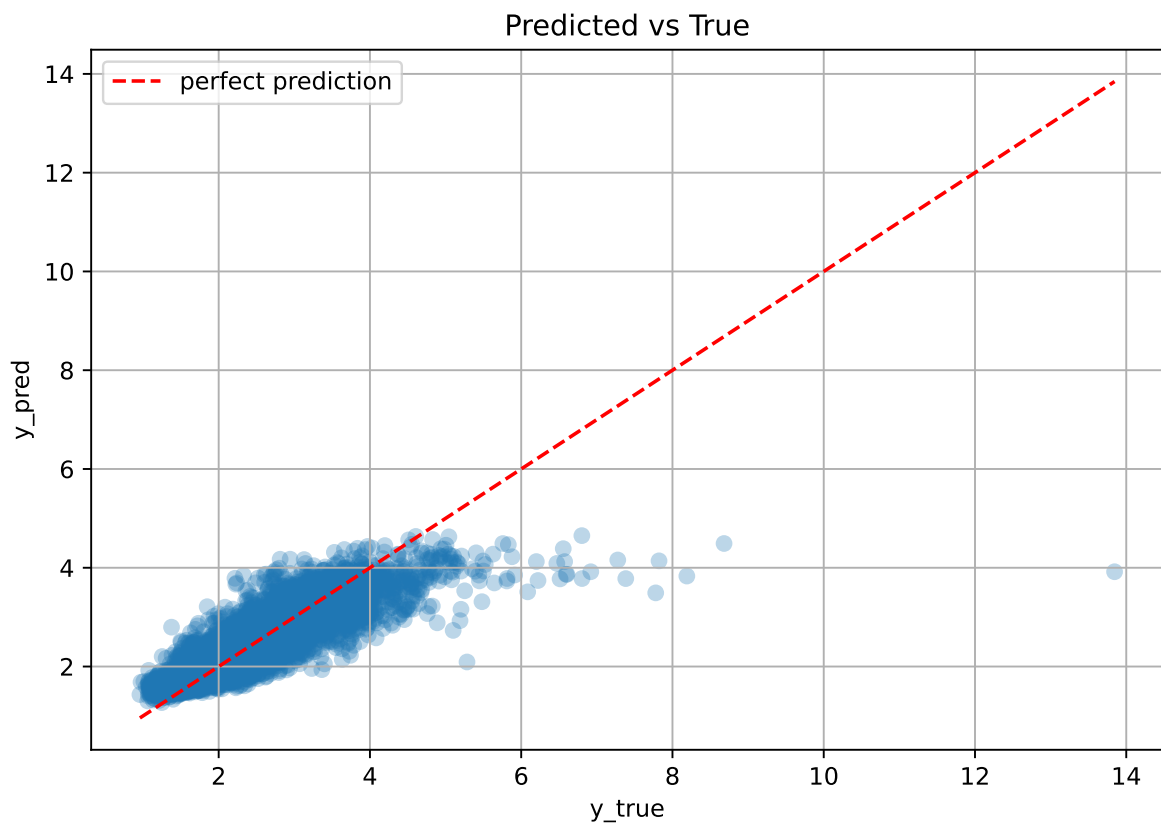
Rank 7: {'reg__n_estimators': 500, 'reg__max_depth': 3, 'reg__learning_rate': 0.027745033904028004, 'reg__num_leaves': 45, 'reg__subsample': 0.696664599264767, 'reg__colsample_bytree': 0.5385272159103219, 'reg__min_child_samples': 36, 'reg__reg_alpha': 4.5061867719861795, 'reg__reg_lambda': 15.622968301245022, 'reg__min_split_gain': 0.6691387336146313, 'reg__path_smooth': 1.64416372900768, 'reg__min_child_weight': 0.12513022970142068, 'reg__feature_fraction': 0.6884726910407394, 'reg__bagging_fraction': 0.6729390004681544, 'reg__bagging_freq': 2, 'reg__max_bin': 216, 'reg__min_data_per_group': 178}

Rank 8: {'reg__n_estimators': 700, 'reg__max_depth': 5, 'reg__learning_rate': 0.020565329961208852, 'reg__num_leaves': 50, 'reg__subsample': 0.6555797399429402, 'reg__colsample_bytree': 0.7522668389859193, 'reg__min_child_samples': 47, 'reg__reg_alpha': 6.842550540011511, 'reg__reg_lambda': 18.609965052239755, 'reg__min_split_gain': 0.7481322771079134, 'reg__path_smooth': 2.813916410677085, 'reg__min_child_weight': 0.05422716330836655, 'reg__feature_fraction': 0.5244842528558655, 'reg__bagging_fraction': 0.7063540494981582, 'reg__bagging_freq': 2, 'reg__max_bin': 230, 'reg__min_data_per_group': 174}

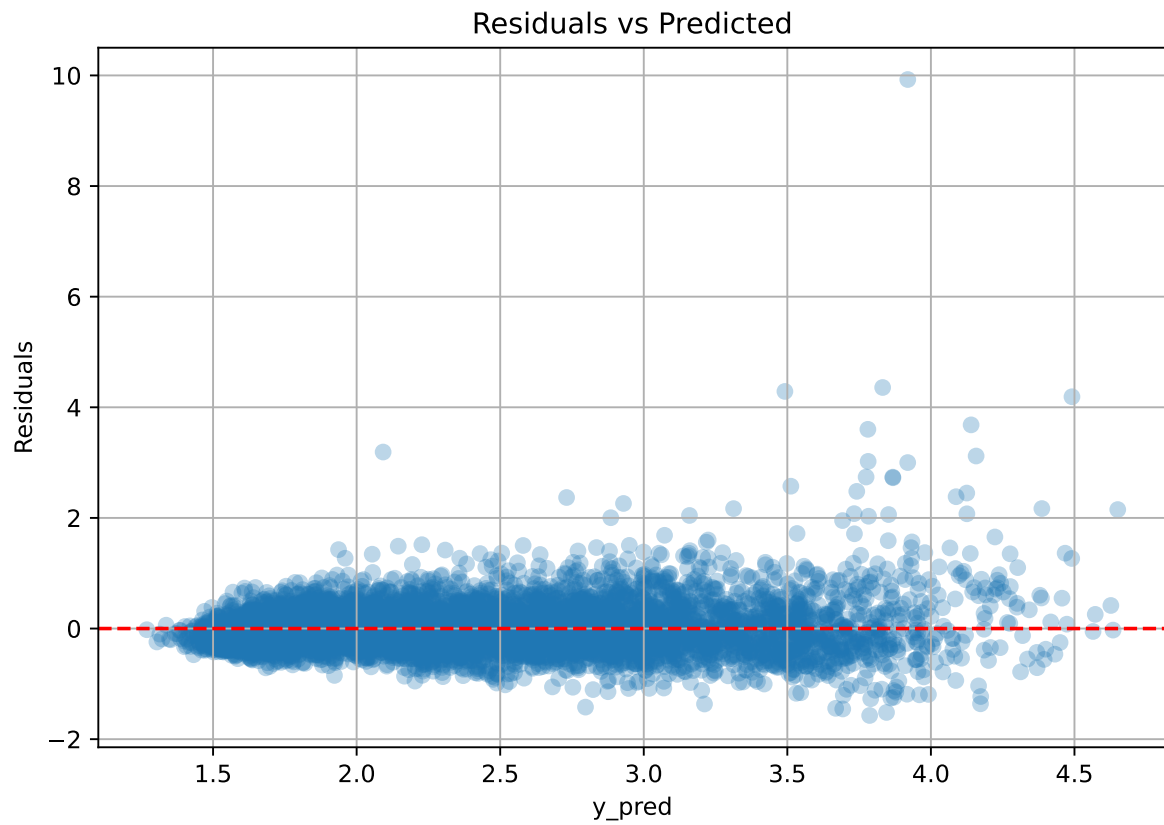
Rank 9: {'reg__n_estimators': 700, 'reg__max_depth': 5, 'reg__learning_rate': 0.017879004800802917, 'reg__num_leaves': 54, 'reg__subsample': 0.865800079152772, 'reg__colsample_bytree': 0.8153804193678105, 'reg__min_child_samples': 50, 'reg__reg_alpha': 8.400803835119325, 'reg__reg_lambda': 12.35617845232225, 'reg__min_split_gain': 0.6879978135844491, 'reg__path_smooth': 3.166673964408216, 'reg__min_child_weight': 0.13152140843547647, 'reg__feature_fraction': 0.5628213973280431, 'reg__bagging_fraction': 0.7219011587518704, 'reg__bagging_freq': 2, 'reg__max_bin': 195, 'reg__min_data_per_group': 183}

Rank 10: {'reg__n_estimators': 700, 'reg__max_depth': 5, 'reg__learning_rate': 0.02102100275759144, 'reg__num_leaves': 50, 'reg__subsample': 0.8131680027345664, 'reg__colsample_bytree': 0.5737177690766131, 'reg__min_child_samples': 37, 'reg__reg_alpha': 7.524258165216643, 'reg__reg_lambda': 25.009543624038095, 'reg__min_split_gain': 0.8245108170775548, 'reg__path_smooth': 1.286507383007297, 'reg__min_child_weight': 0.04894071553688423, 'reg__feature_fraction': 0.6436642121933864, 'reg__bagging_fraction': 0.7016568072904169, 'reg__bagging_freq': 2, 'reg__max_bin': 221, 'reg__min_data_per_group': 176}

3.5.2 Scatter



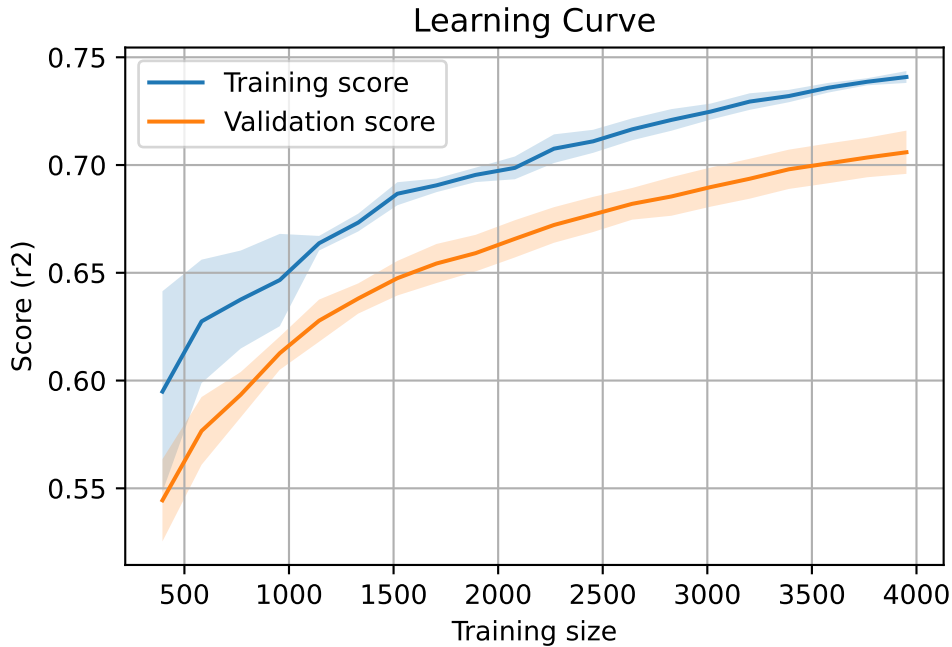
3.5.3 Scatter residuals



3.5.4 Learning curve

`Len(group)` : 8068, `Len(training)` : 5648,

`Len(training_real)` : 3954, `Len(test_of_training)` : 1694, `Len(test)` : 2420



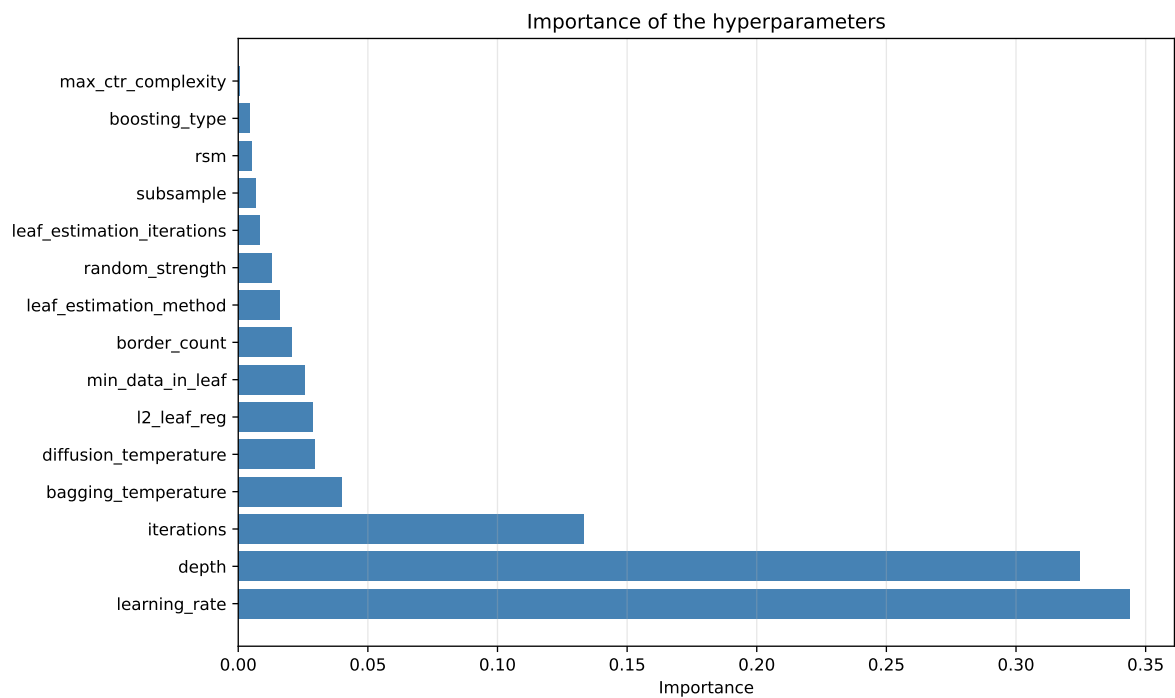
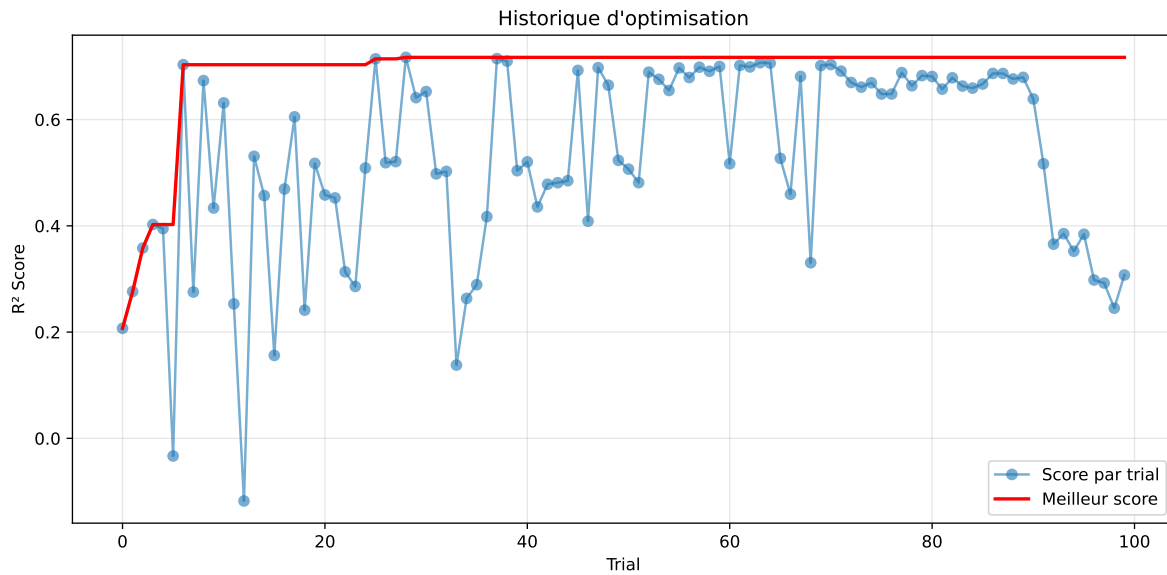
3.6 Model : CatBoost

3.6.1 Gridsearch

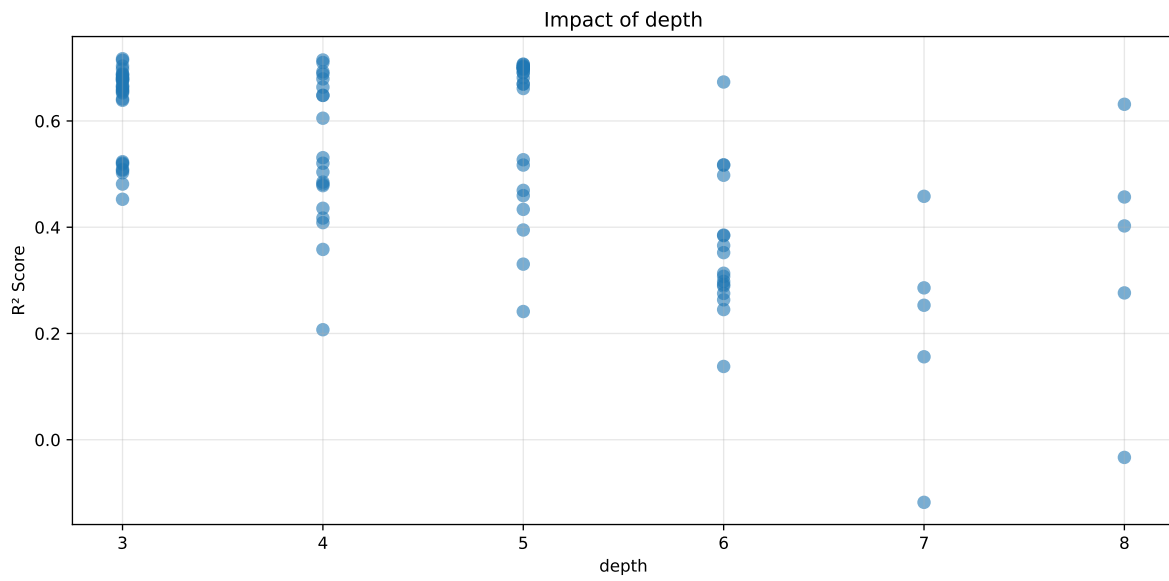
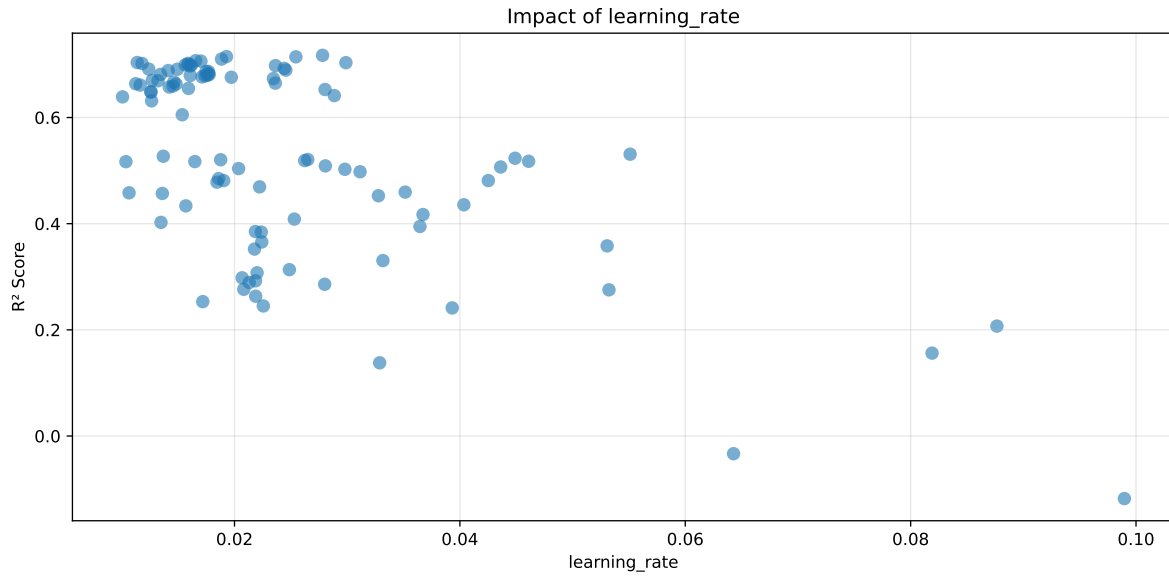
Distribution y_train vs y_test: y_train: mean=2.347, std=0.801, min=1.041, max=8.682
y_test: mean=2.362, std=0.789, min=0.959, max=13.845

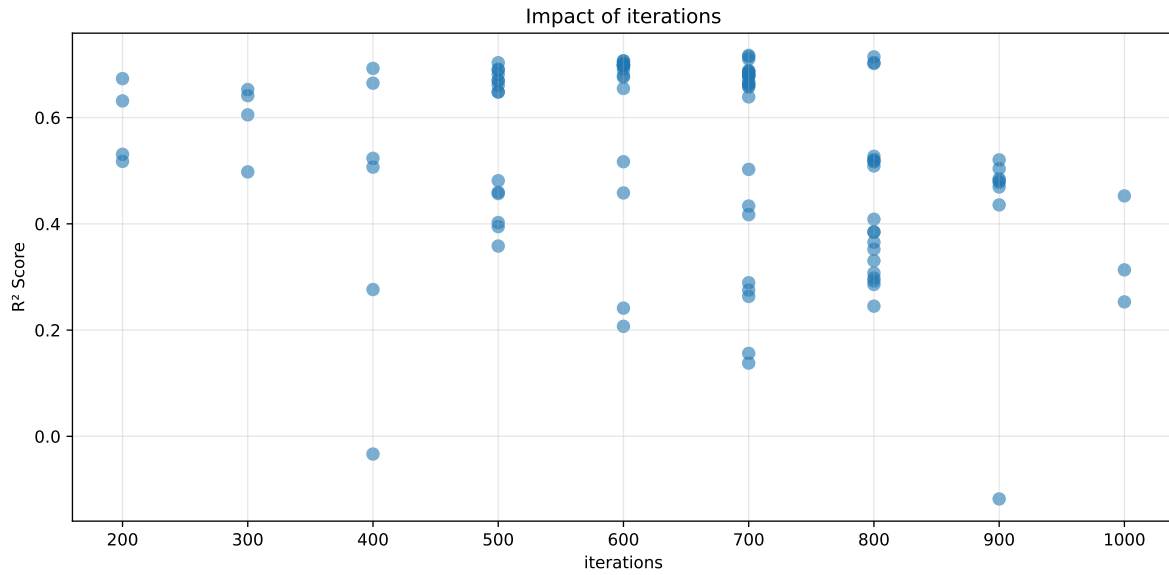
===== TOP Importance FEATURES =====

feature importance	r_temperature_st_3h_slope	r_umidity	r_CO2
24.419716	18.813616	10.035570	
r_temperature_st_3h_diff_last 7.627774	r_NH3 7.603459	r_temperature 5.988372	
r_NH3_st_3h_median 5.898798	r_NH3_st_3h_std 4.090257	r_umidity_st_3h_std 3.417122	
r_temperature_st_3h_mean 3.294466	r_NH3_st_3h_slope 3.272896	r_THI 2.398938	
r_umidity_st_3h_diff_last 2.245291	r_NH3_st_3h_diff_last 0.893726		



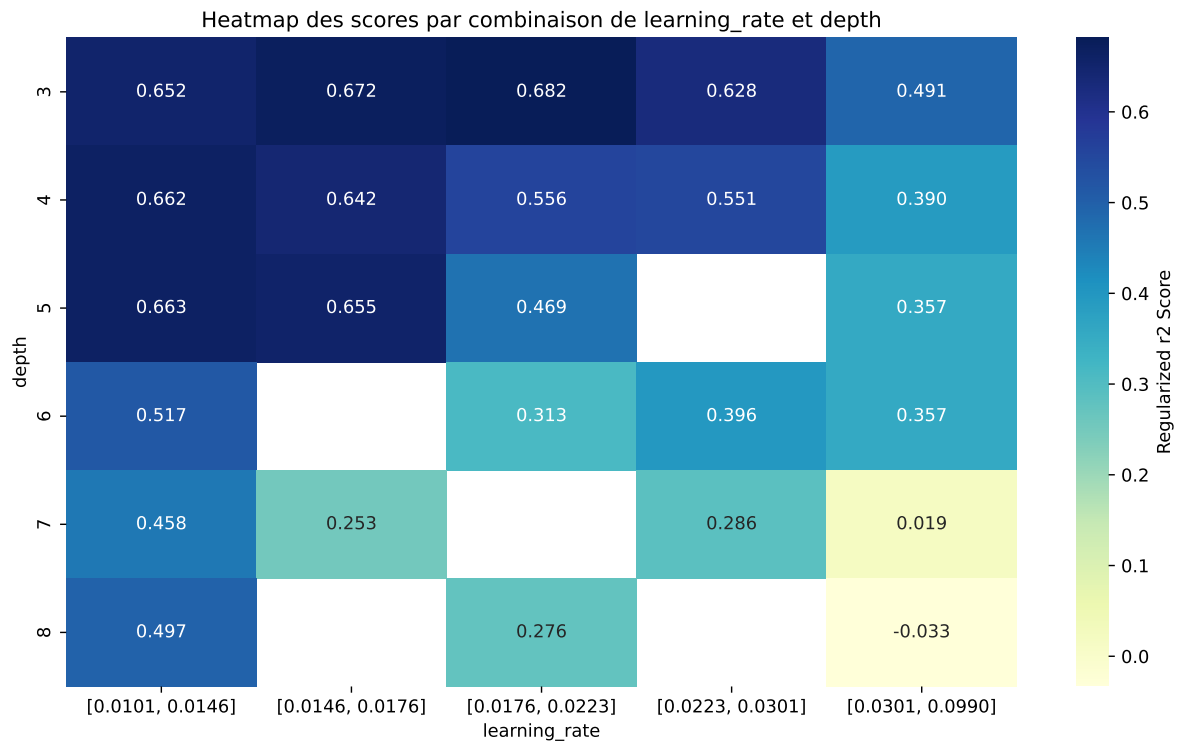
Paramètres avec >5% d'importance : ['learning_rate', 'depth', 'iterations']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7528	0.7172	0.6656	0.0073	0.7172	0.6688	-0.0483	0.9326	1.0498
2	0.7494	0.7148	0.666	0.0079	0.7148	0.6689	-0.0459	0.9358	1.0484
3	0.7464	0.7143	0.6661	0.0084	0.7143	0.6682	-0.0461	0.9355	1.0448
4	0.7432	0.71	0.6614	0.0081	0.71	0.6659	-0.0441	0.9379	1.0468
5	0.7401	0.707	0.6603	0.0072	0.707	0.6662	-0.0409	0.9422	1.0468
6	0.7384	0.7063	0.6599	0.0068	0.7063	0.6664	-0.0399	0.9436	1.0455
7	0.7367	0.7036	0.6576	0.0068	0.7036	0.6634	-0.0402	0.9429	1.047
8	0.7383	0.7033	0.6541	0.0085	0.7033	0.6618	-0.0415	0.941	1.0497
9	0.7357	0.702	0.6569	0.0074	0.702	0.6641	-0.0379	0.946	1.0479
10	0.7332	0.7018	0.655	0.0061	0.7018	0.6612	-0.0406	0.9421	1.0448
---	---	---	---	---	---	---	---	---	---
100	0.9877	0.8035	0.7495	0.0068	-0.1179	0.7648	-0.0386	0.9519	1.2293

Hyperparameters:

Rank 1: {'reg__iterations': 700, 'reg__depth': 3, 'reg__learning_rate': 0.027815605066969656, 'reg__l2_leaf_reg': 22.015932452138163, 'reg__bagging_temperature': 3.2383944635445654, 'reg__random_strength': 4.237537473664137, 'reg__subsample': 0.5039232798045266, 'reg__min_data_in_leaf': 34, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.6567758464692824, 'reg__border_count': 48, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 6581.568234942243}

Rank 2: {'reg__iterations': 700, 'reg__depth': 4, 'reg__learning_rate': 0.019288120119489438, 'reg__l2_leaf_reg': 8.631946950699964, 'reg__bagging_temperature': 3.502819810677237, 'reg__random_strength': 4.217234187315434, 'reg__subsample': 0.8447785904519688, 'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.6875331473360468, 'reg__border_count': 63, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 5300.537770112683}

Rank 3: {'reg__iterations': 800, 'reg__depth': 3, 'reg__learning_rate': 0.02544734850281001, 'reg__l2_leaf_reg': 20.802191314859236, 'reg__bagging_temperature': 2.9511197927395227, 'reg__random_strength': 4.563028957864522, 'reg__subsample': 0.6855252647084367, 'reg__min_data_in_leaf': 34, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.6490599553719526, 'reg__border_count': 48, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 6926.5739555609025}

Rank 4: {'reg__iterations': 700, 'reg__depth': 4, 'reg__learning_rate': 0.018846943949449694, 'reg__l2_leaf_reg': 18.17778740637293, 'reg__bagging_temperature': 2.565756261793678, 'reg__random_strength': 4.231570858479117, 'reg__subsample': 0.8263538305693254, 'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 5, 'reg__rsm':

0.6872251832923392, 'reg__border_count': 64, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 5368.21891330839}

Rank 5: {'reg__iterations': 600, 'reg__depth': 5, 'reg__learning_rate': 0.016551708213303228, 'reg__l2_leaf_reg': 14.087307139191392, 'reg__bagging_temperature': 3.0065124603961344, 'reg__random_strength': 3.3612215923831994, 'reg__subsample': 0.8753666308855516, 'reg__min_data_in_leaf': 12, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.7245173517432292, 'reg__border_count': 48, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 6456.090571639114}

Rank 6: {'reg__iterations': 600, 'reg__depth': 5, 'reg__learning_rate': 0.01701976905716808, 'reg__l2_leaf_reg': 13.95935492482247, 'reg__bagging_temperature': 3.065994600413367, 'reg__random_strength': 3.7768376094600766, 'reg__subsample': 0.5154655427429365, 'reg__min_data_in_leaf': 20, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.760507342866732, 'reg__border_count': 51, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 6601.07396461943}

Rank 7: {'reg__iterations': 800, 'reg__depth': 5, 'reg__learning_rate': 0.011367616604631594, 'reg__l2_leaf_reg': 4.125856594037563, 'reg__bagging_temperature': 3.0247895164440837, 'reg__random_strength': 3.2967266004090217, 'reg__subsample': 0.8801870245710517, 'reg__min_data_in_leaf': 37, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.7827524935587269, 'reg__border_count': 86, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 4335.142374546589}

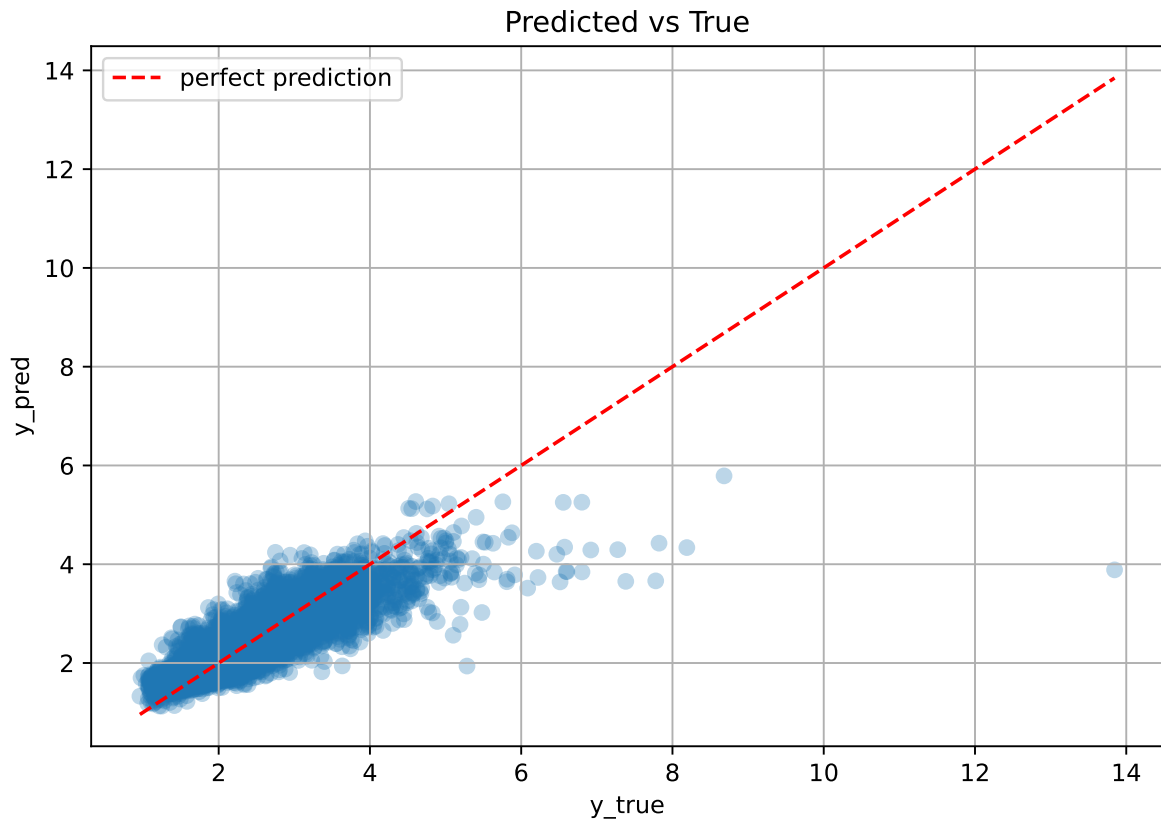
Rank 8: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.02988468261092555, 'reg__l2_leaf_reg': 15.439866825695997, 'reg__bagging_temperature': 2.5346463796805265, 'reg__random_strength': 2.76917302867366, 'reg__subsample': 0.7900790134318896, 'reg__min_data_in_leaf': 31, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.6794872794885113, 'reg__border_count': 101, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 7490.960994118309}

Rank 9: {'reg__iterations': 600, 'reg__depth': 5, 'reg__learning_rate': 0.015871307955540714, 'reg__l2_leaf_reg': 9.74753678663083, 'reg__bagging_temperature': 2.966213328769452, 'reg__random_strength': 3.3710420636163576, 'reg__subsample': 0.8913989048124988, 'reg__min_data_in_leaf': 12, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.6412872874056006, 'reg__border_count': 51, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 4818.326478558295}

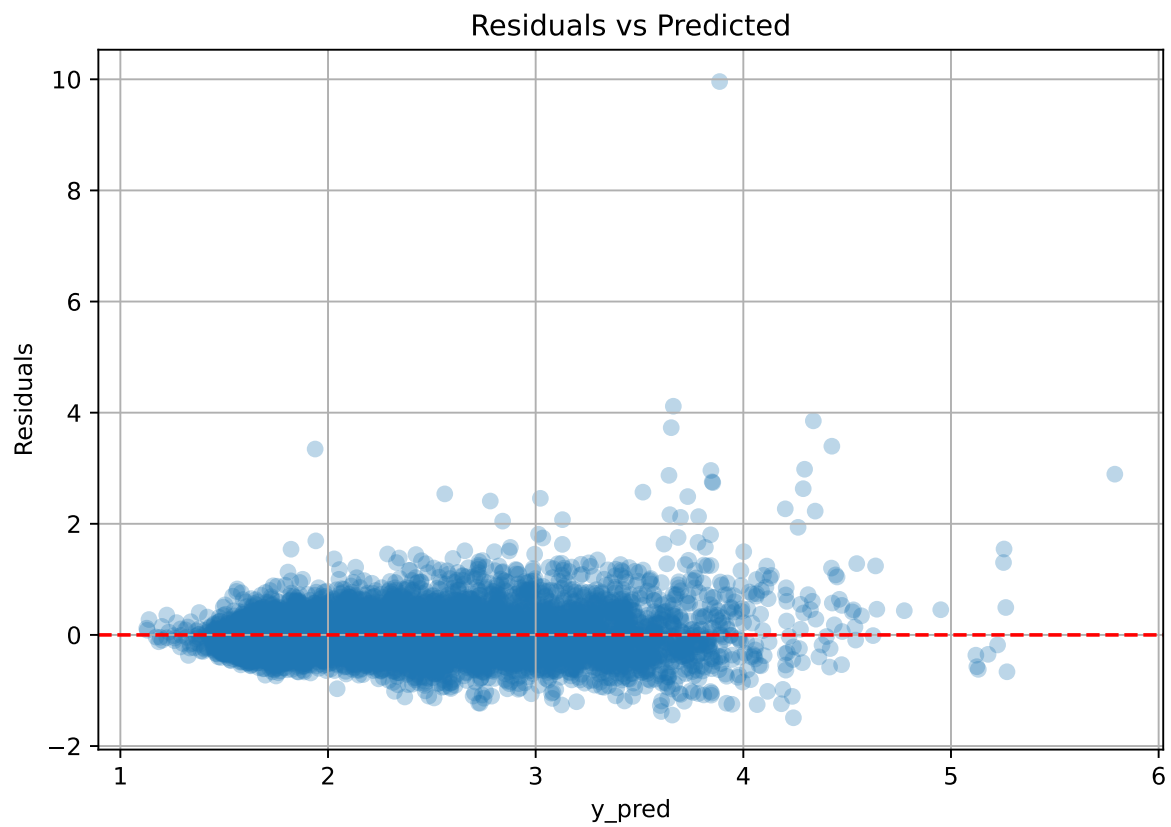
Rank 10: {'reg__iterations': 800, 'reg__depth': 5, 'reg__learning_rate': 0.011793695070736429, 'reg__l2_leaf_reg': 4.7949075265774574, 'reg__bagging_temperature': 4.380278038438814,

```
'reg__random_strength': 3.910503458645914, 'reg__subsample': 0.8765546142009357,  
'reg__min_data_in_leaf': 37, 'reg__leaf_estimation_iterations': 6, 'reg__rsm':  
0.7624461852036151, 'reg__border_count': 47, 'reg__leaf_estimation_method': 'Gradient',  
'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_tempera-  
ture': 4086.7346881920994}
```

3.6.2 Scatter



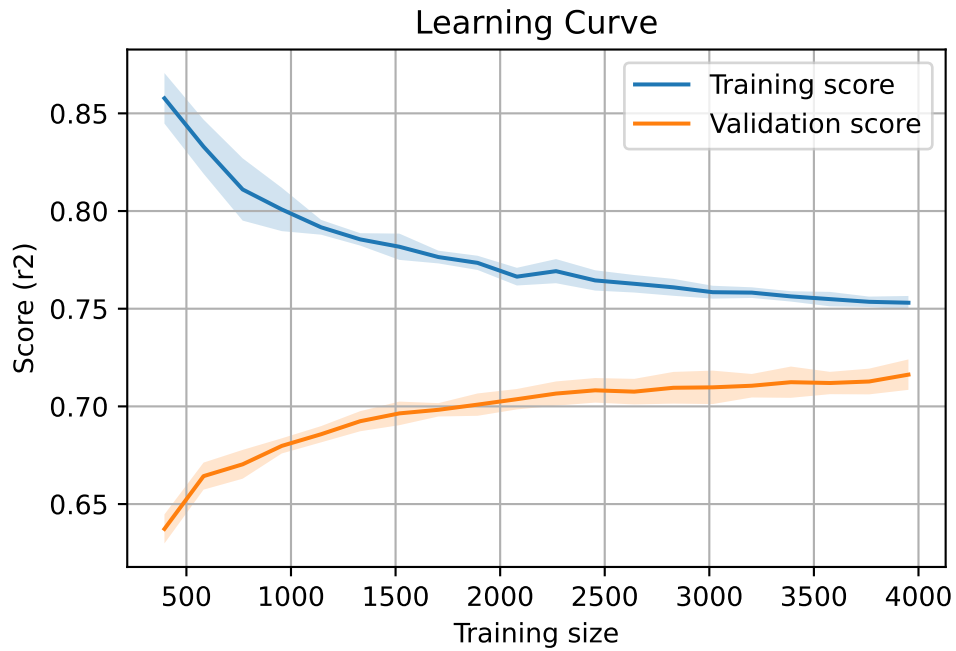
3.6.3 Scatter residuals



3.6.4 Learning curve

`Len(group)` : 8068, `Len(training)` : 5648,

`Len(training_real)` : 3954, `Len(test_of_training)` : 1694, `Len(test)` : 2420



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.556	0.554	0.008	0.506	0.507	-0.048	0.914	1.003	nan
LGBM	0.741	0.706	0.009	0.662	0.68	-0.025	0.964	1.049	0.7058
CatBoost	0.753	0.717	0.007	0.666	0.669	-0.048	0.933	1.05	0.7172

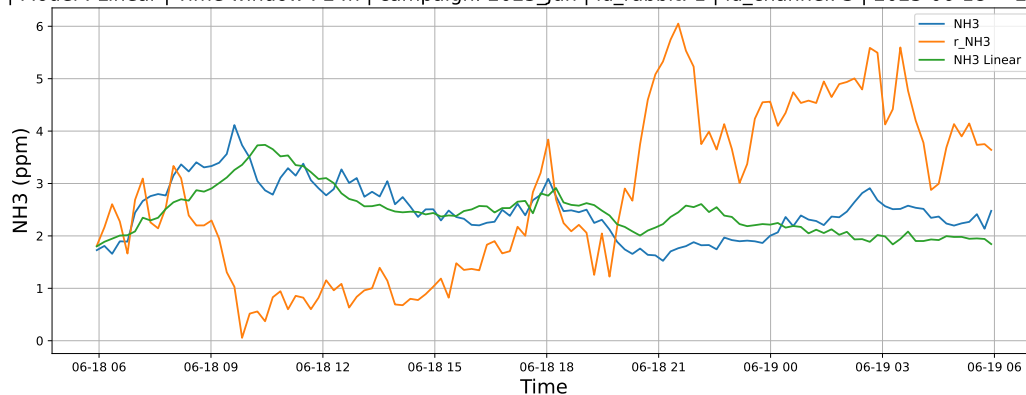
4 Plot

4.1 Standalone plots

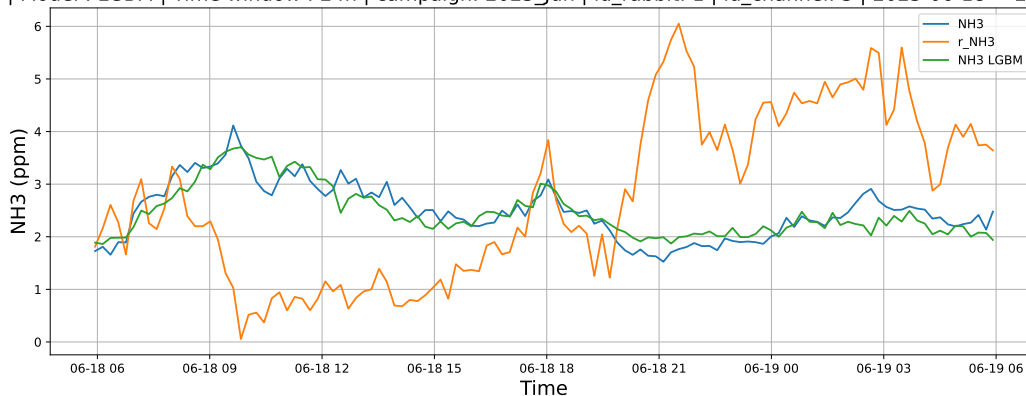
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

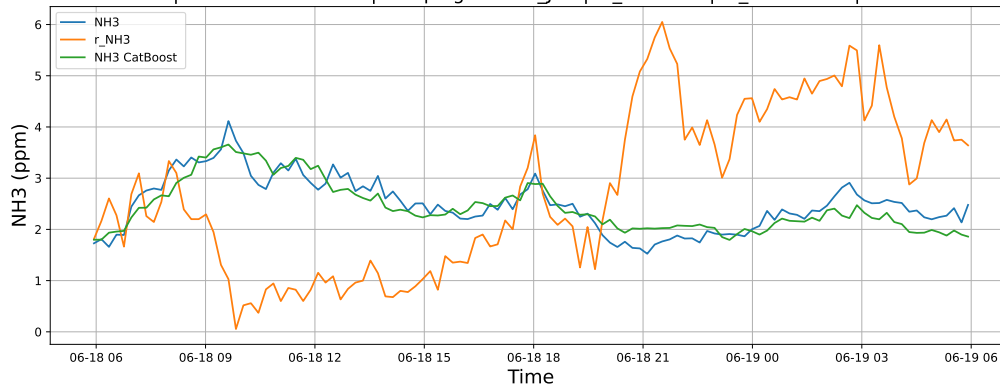
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

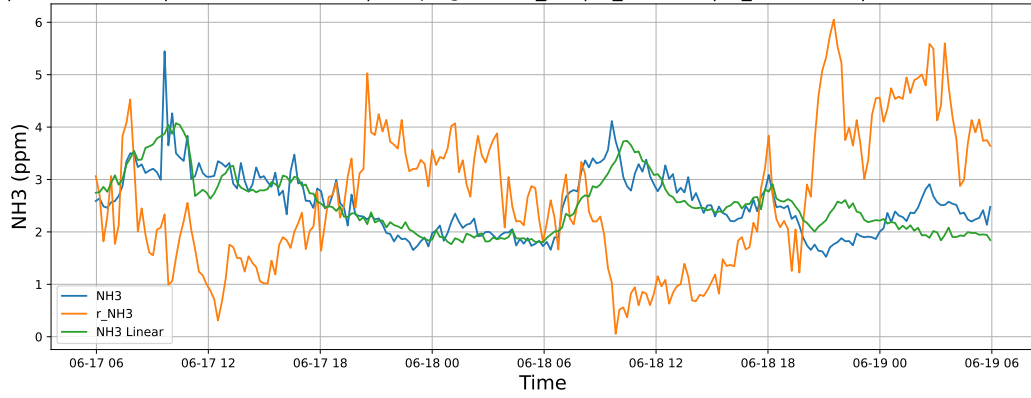


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

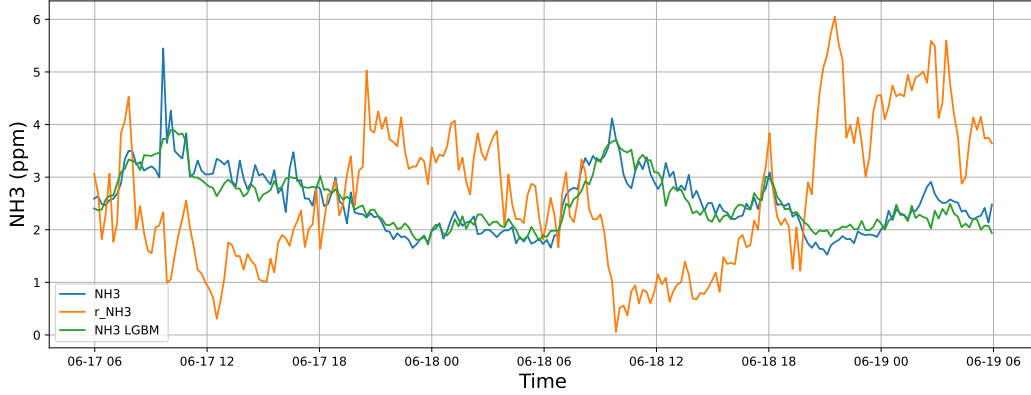


4.1.1.2 Time window : 48

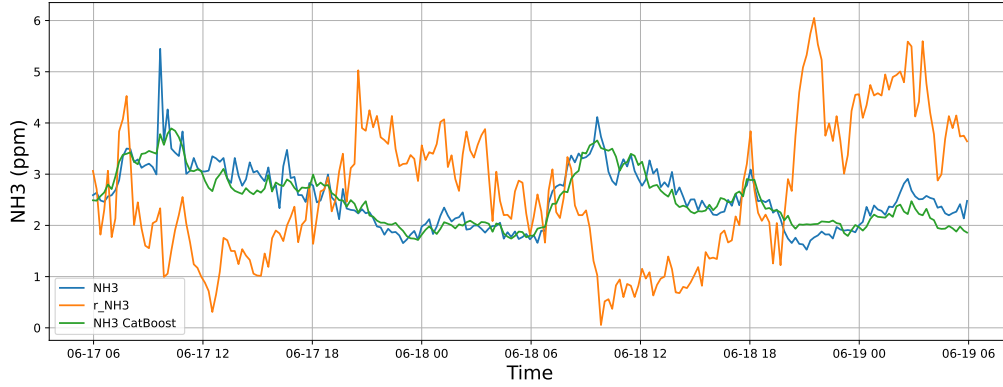
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

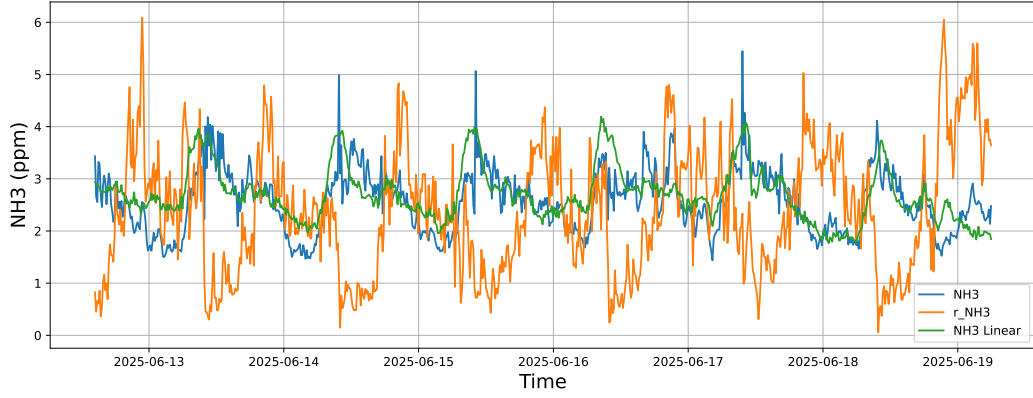


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

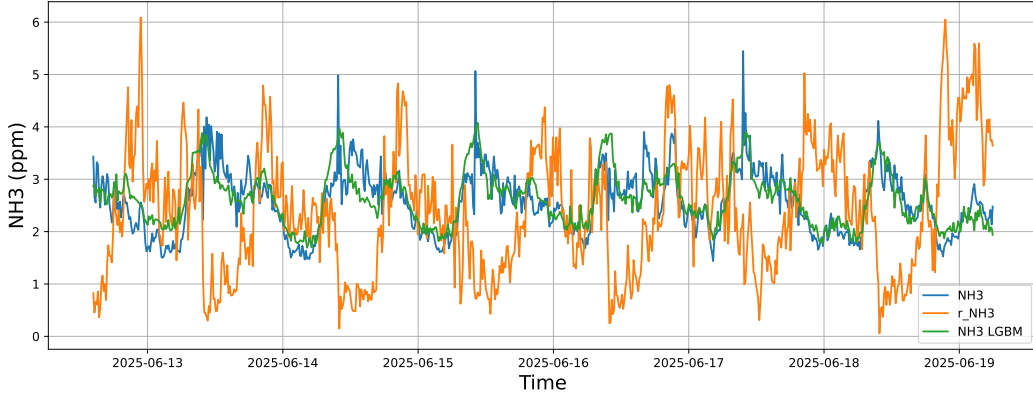


4.1.1.3 Time window : ALL

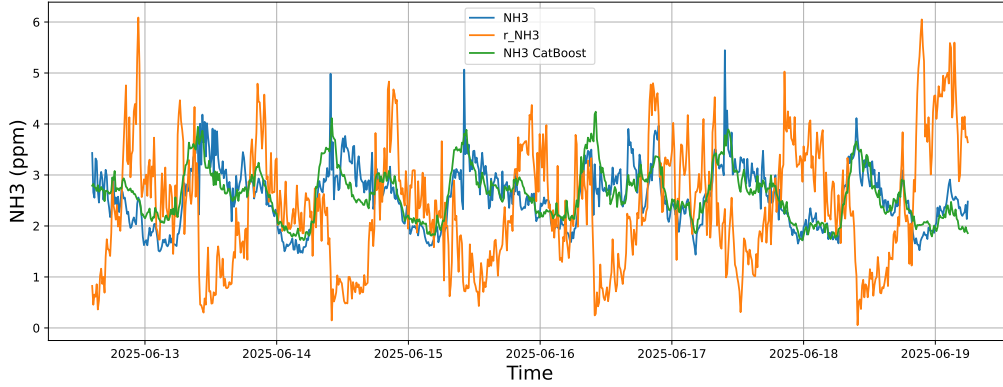
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



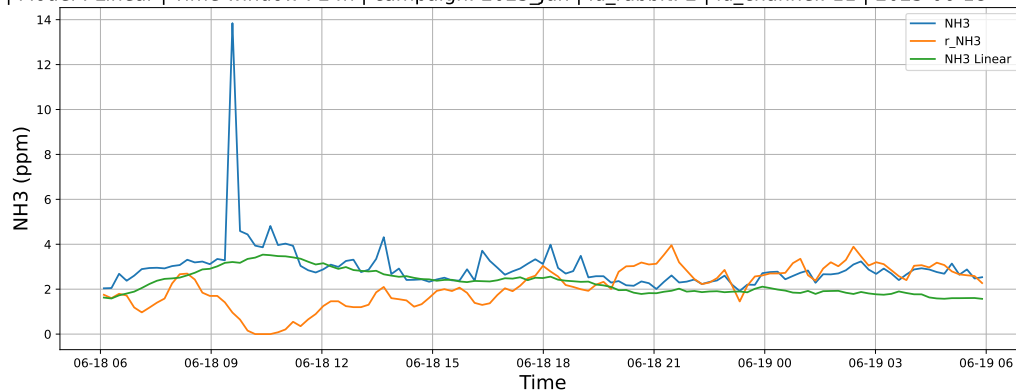
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



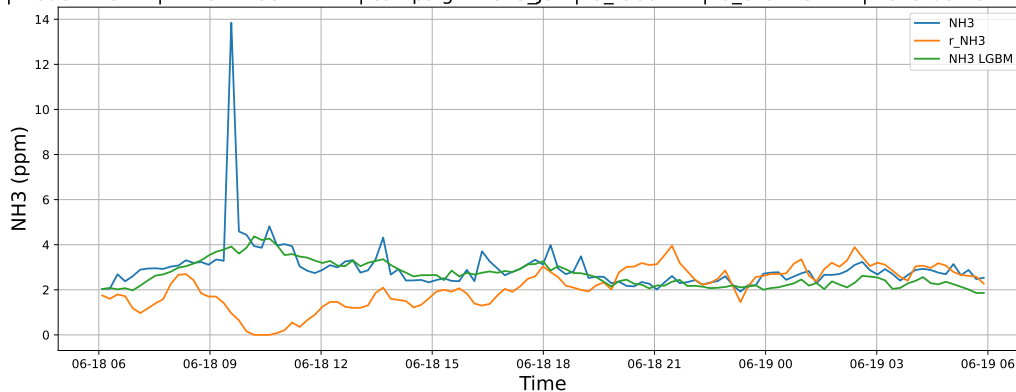
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

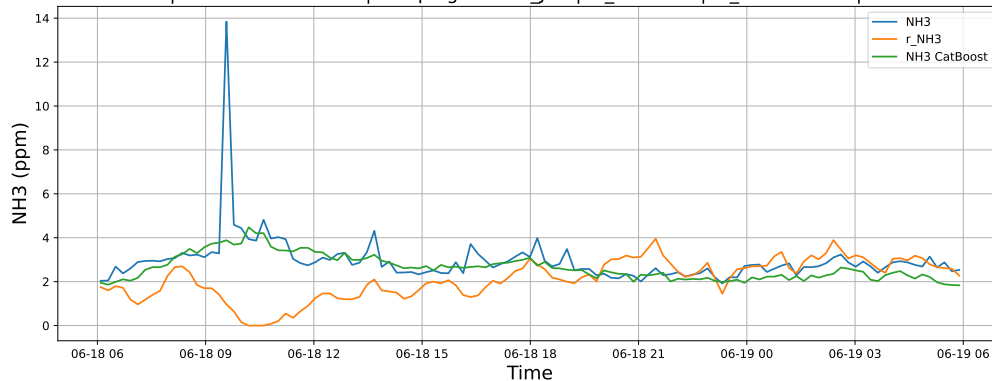
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

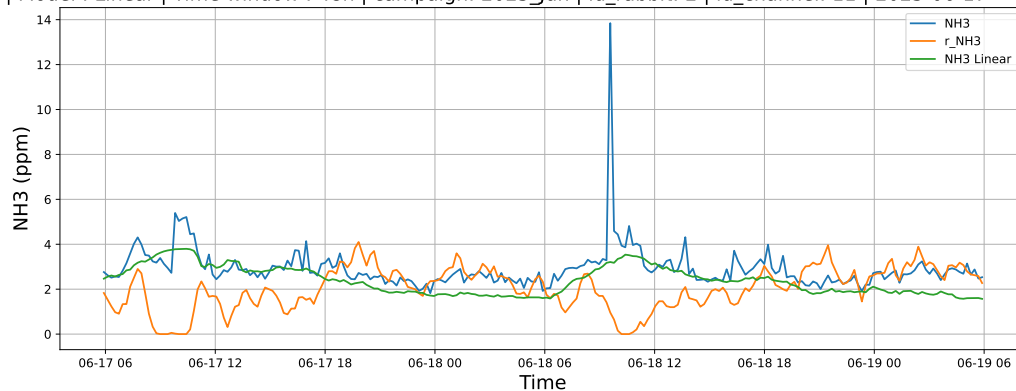


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

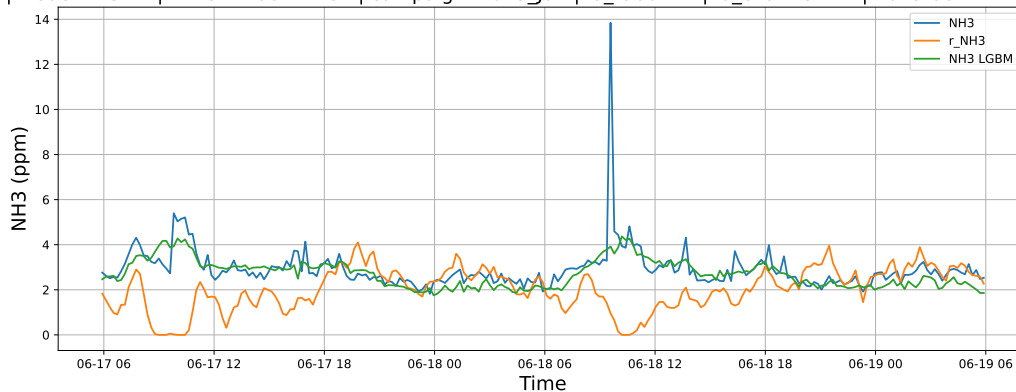


4.1.2.2 Time window : 48

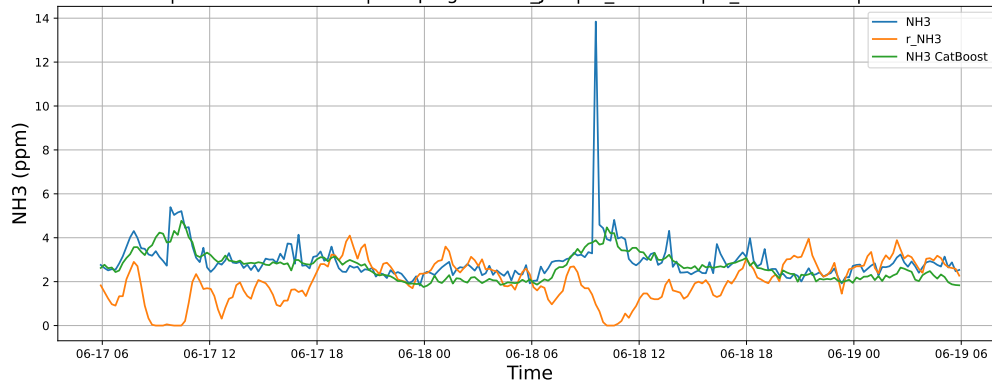
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

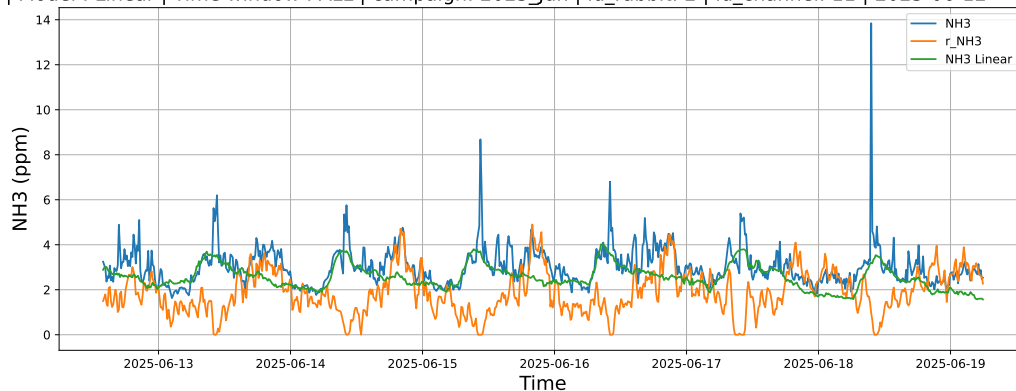


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

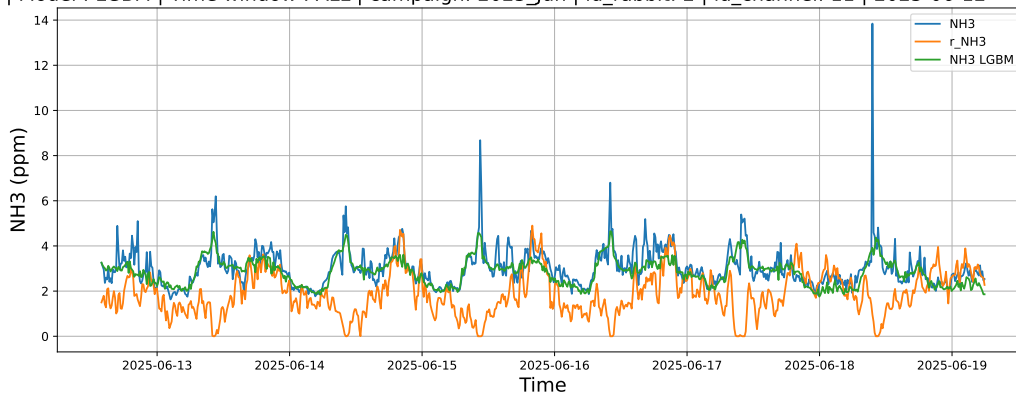


4.1.2.3 Time window : ALL

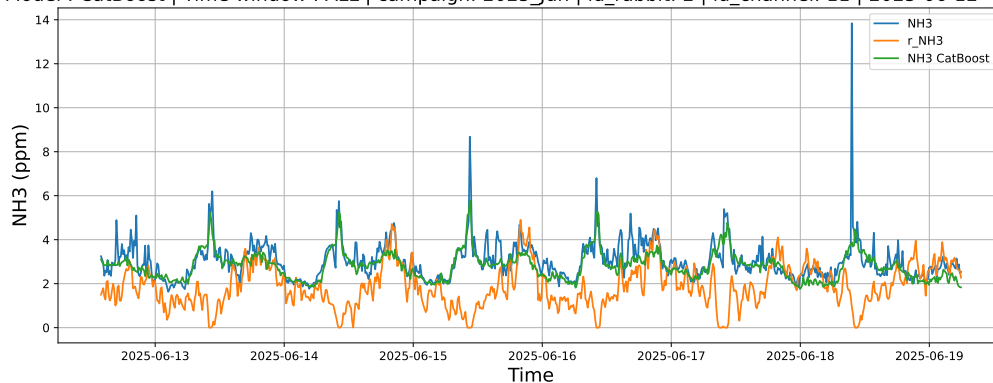
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



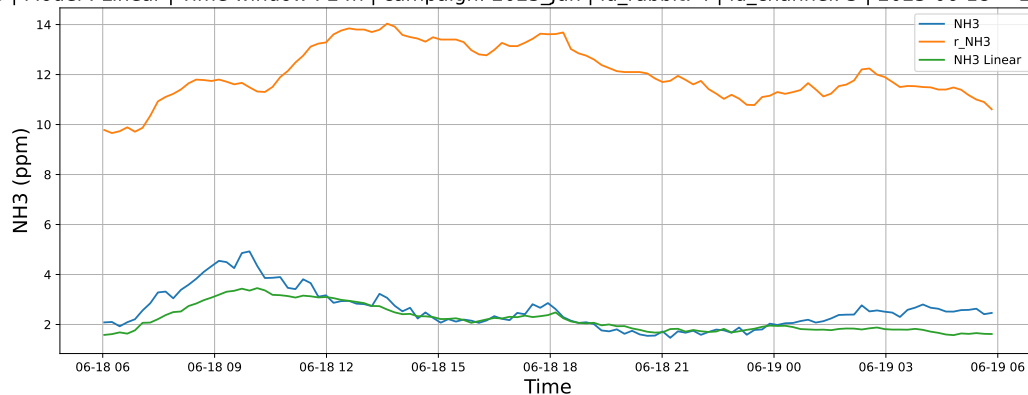
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



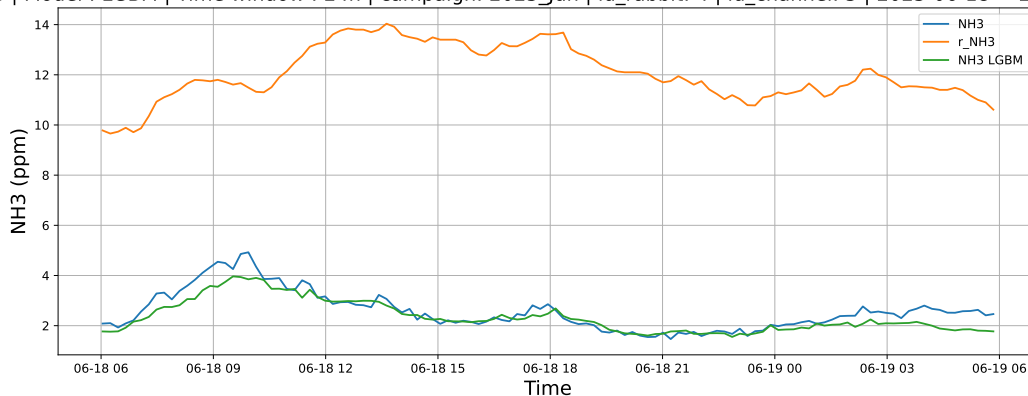
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

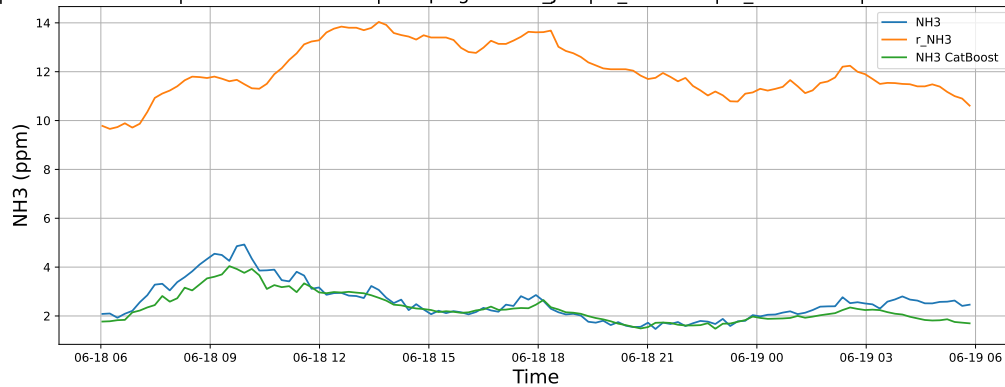
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

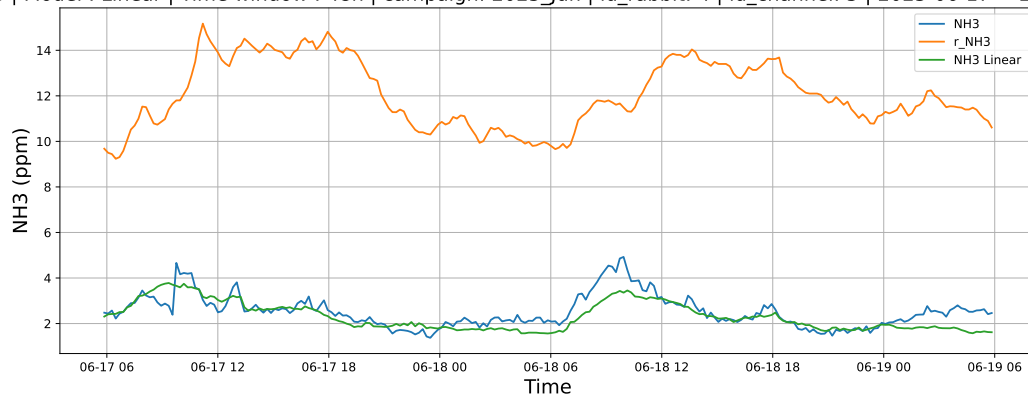


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

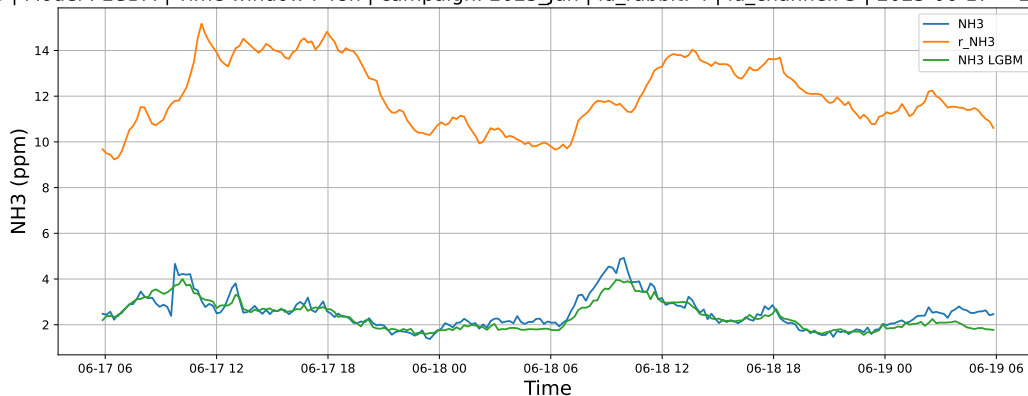


4.1.3.2 Time window : 48

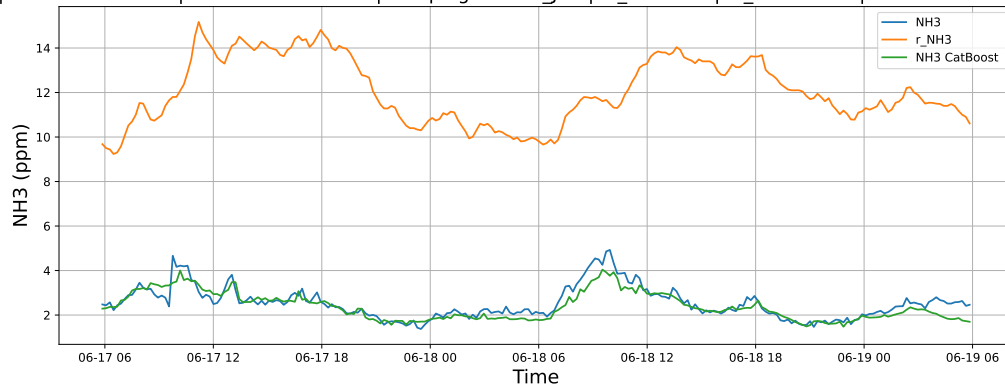
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

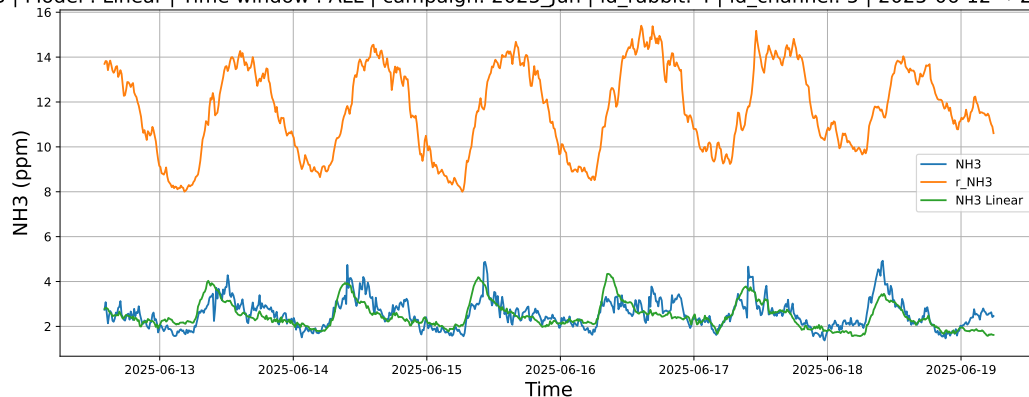


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

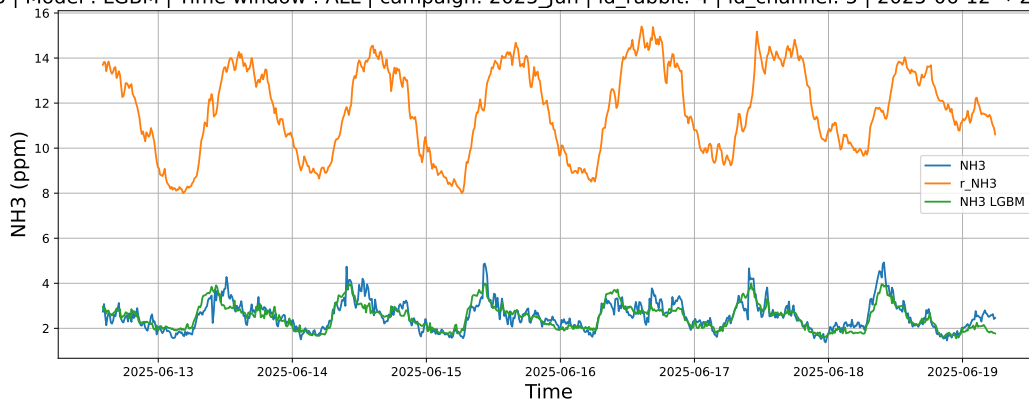


4.1.3.3 Time window : ALL

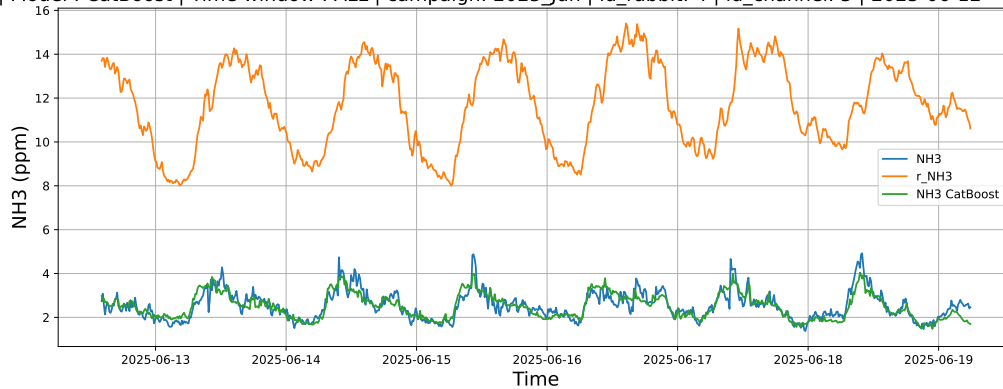
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



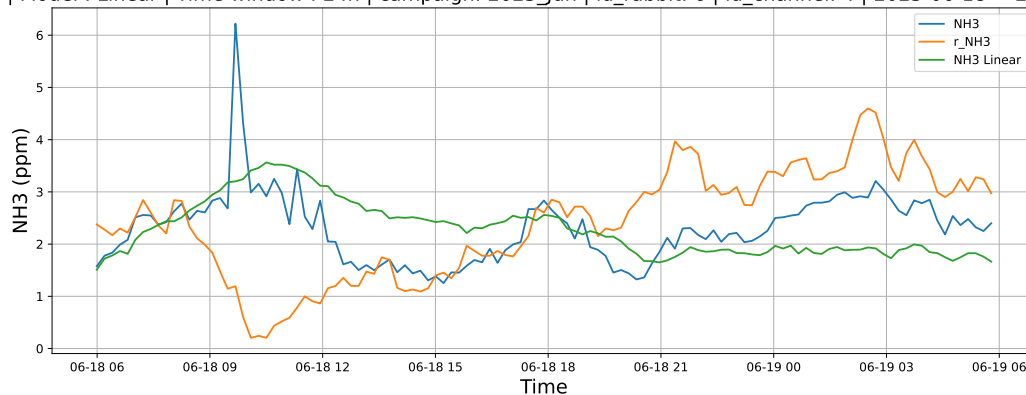
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



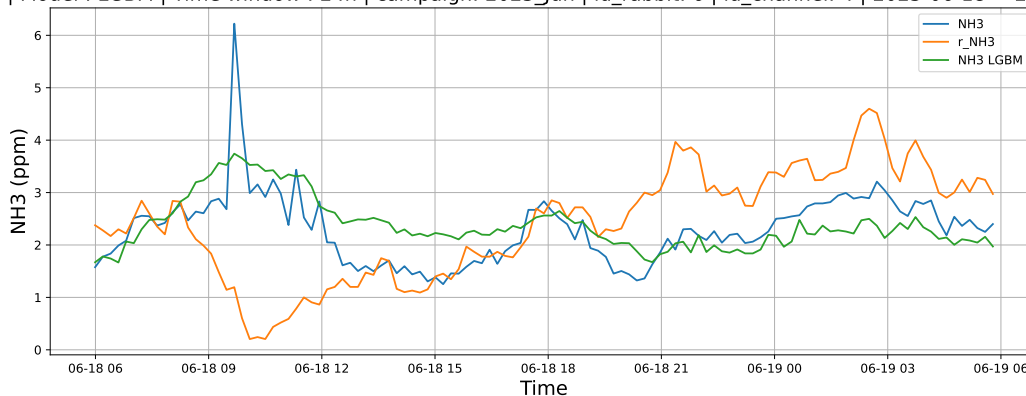
4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

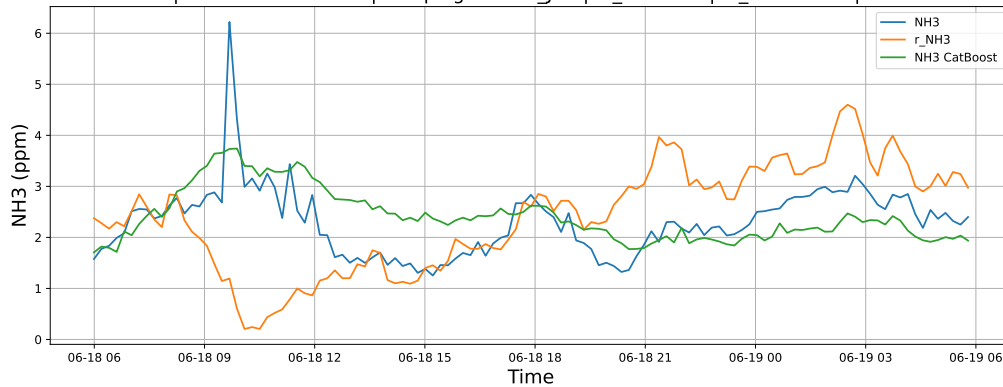
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

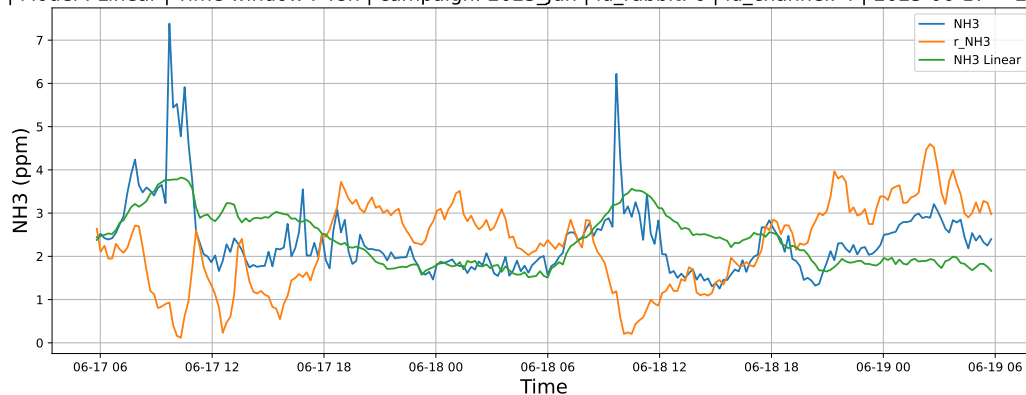


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

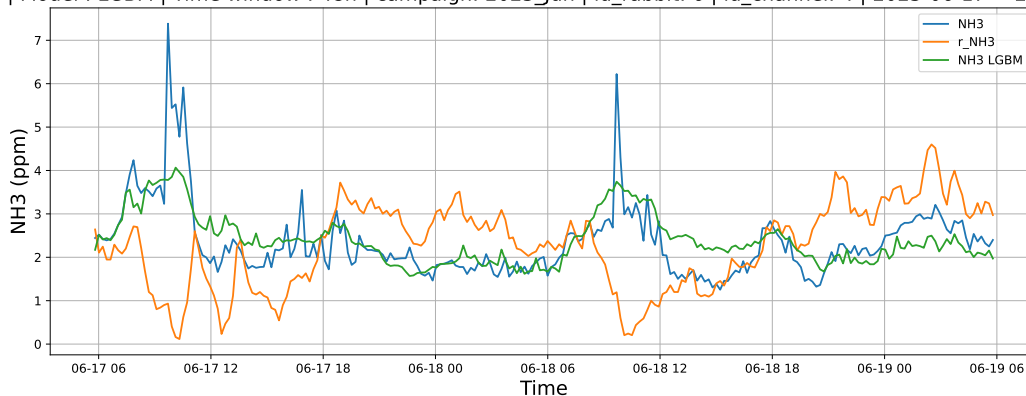


4.1.4.2 Time window : 48

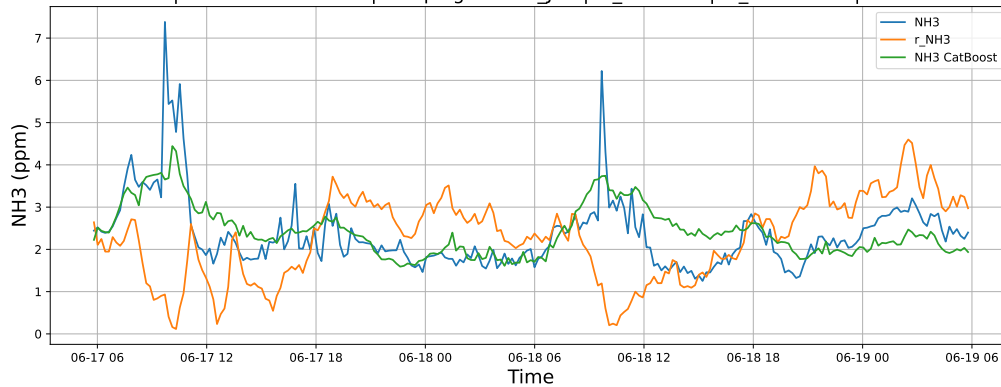
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

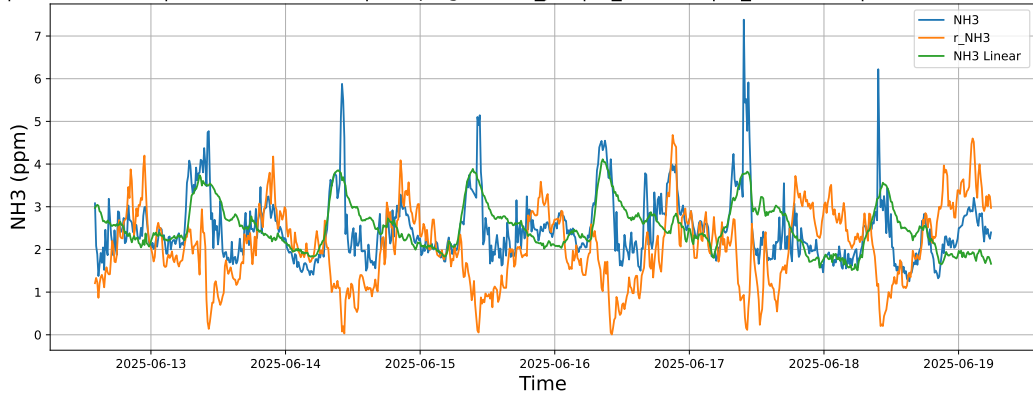


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

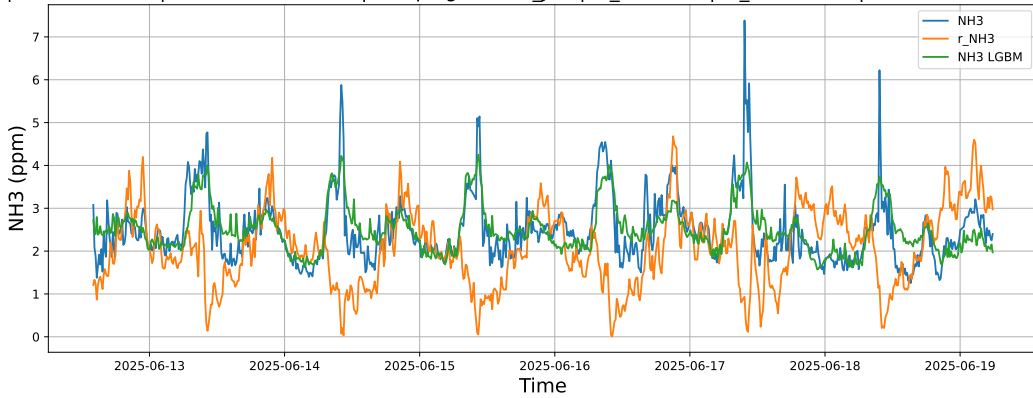


4.1.4.3 Time window : ALL

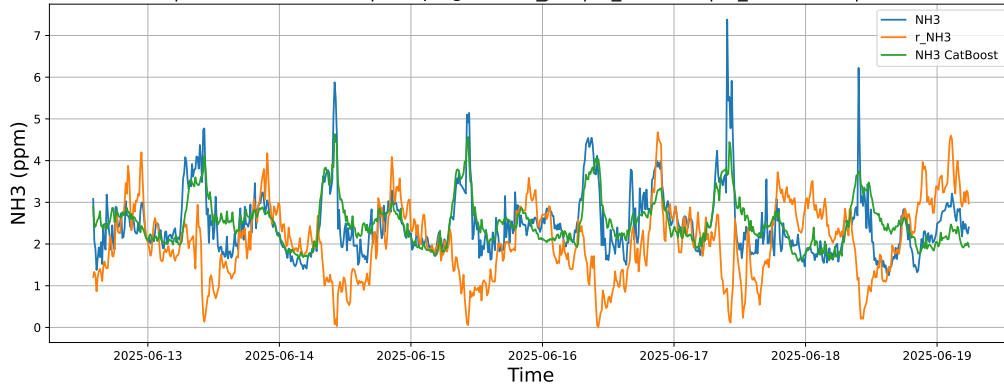
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



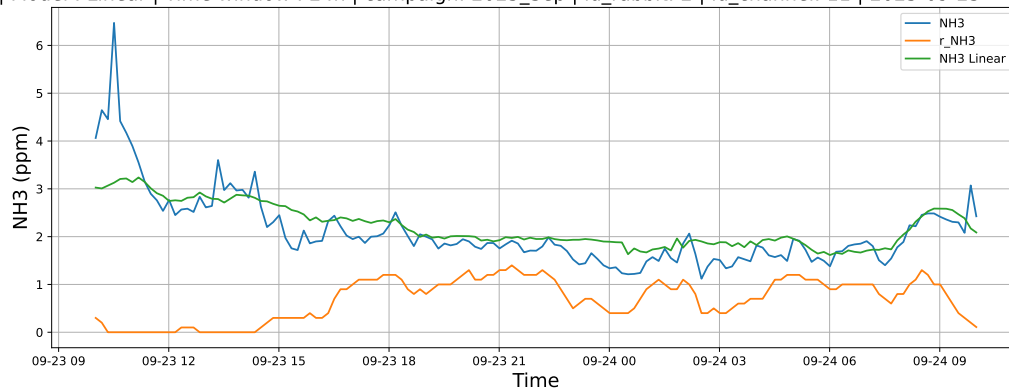
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



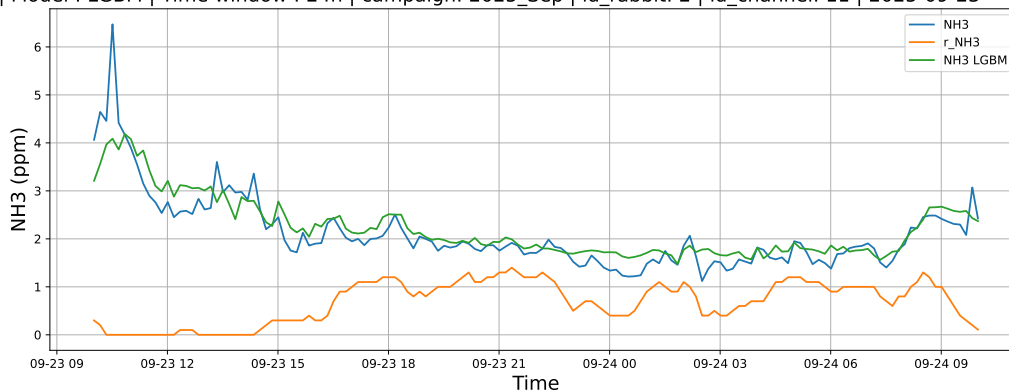
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

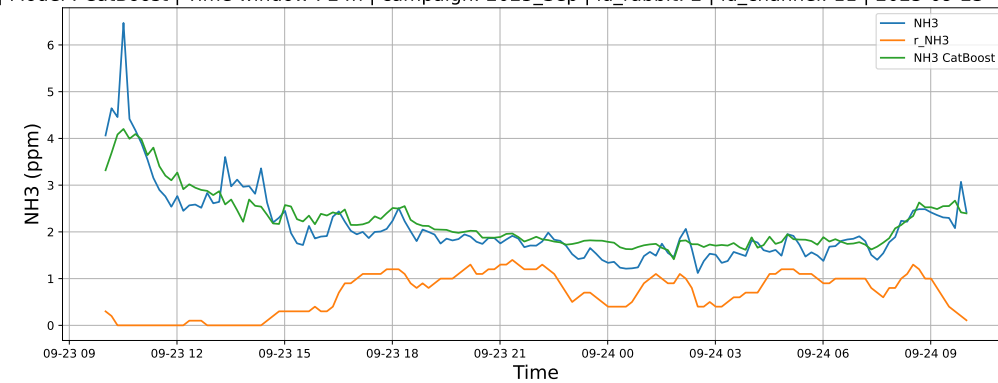
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

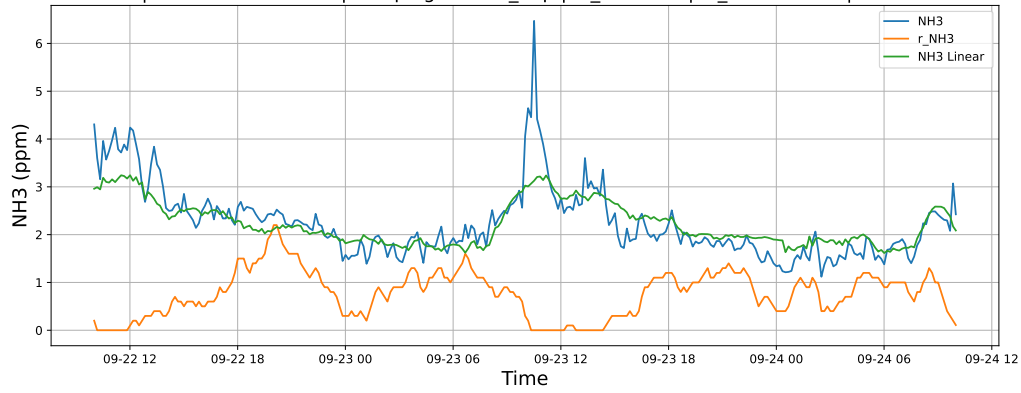


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

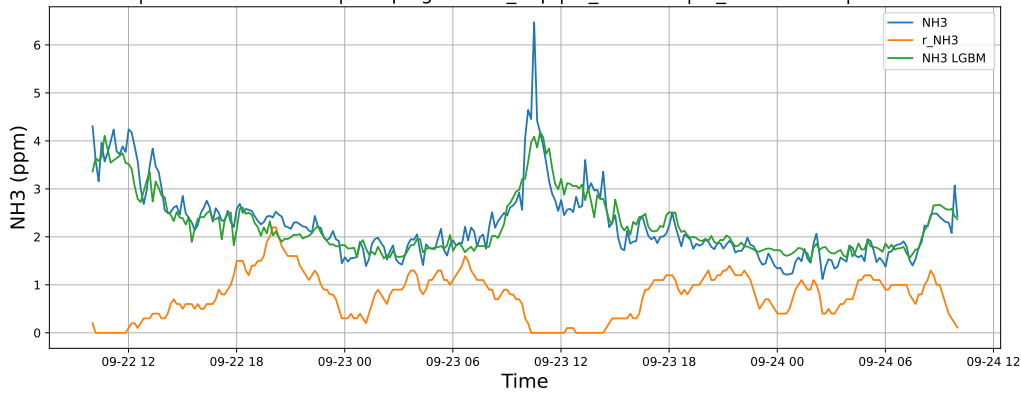


4.1.5.2 Time window : 48

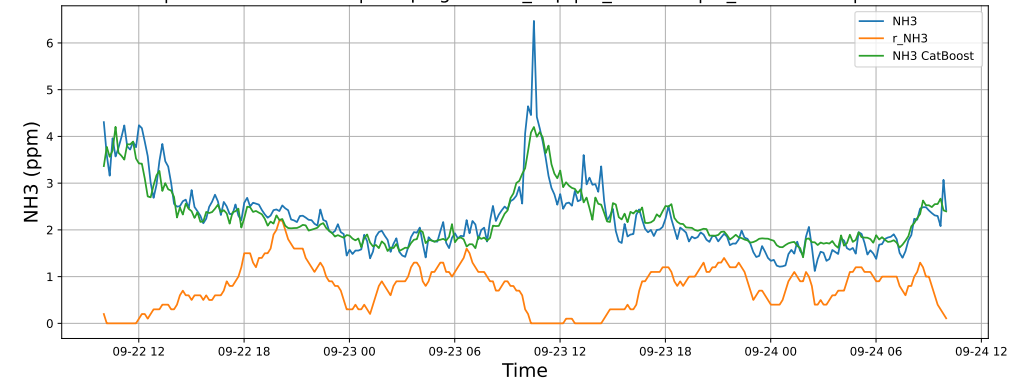
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

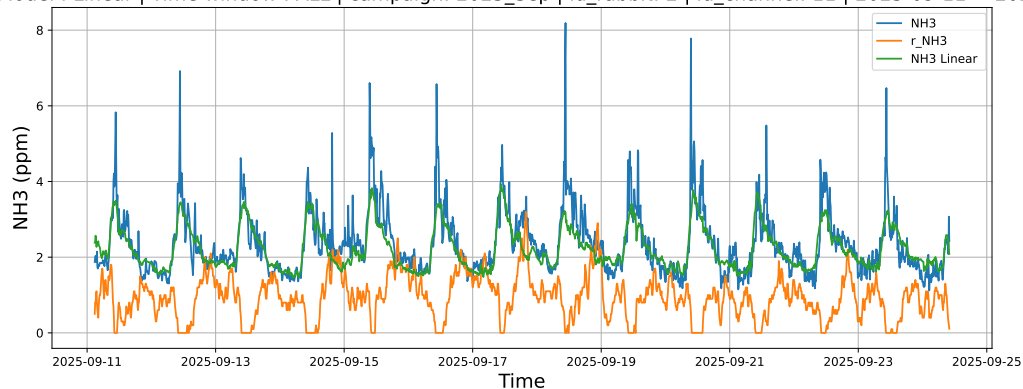


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

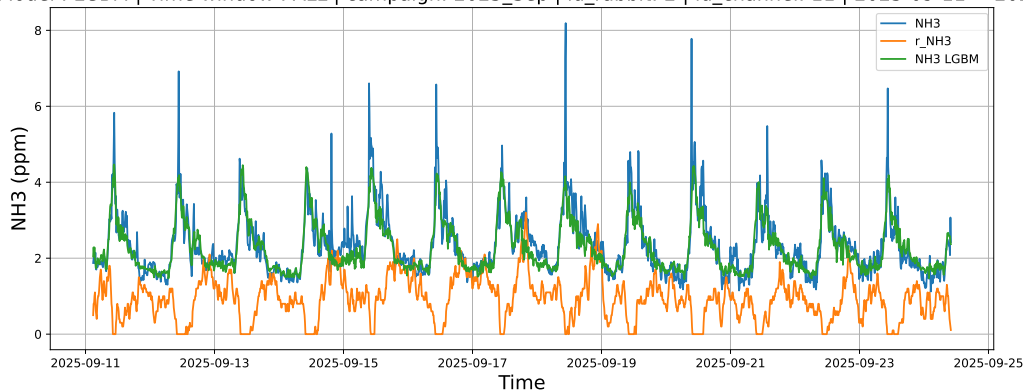


4.1.5.3 Time window : ALL

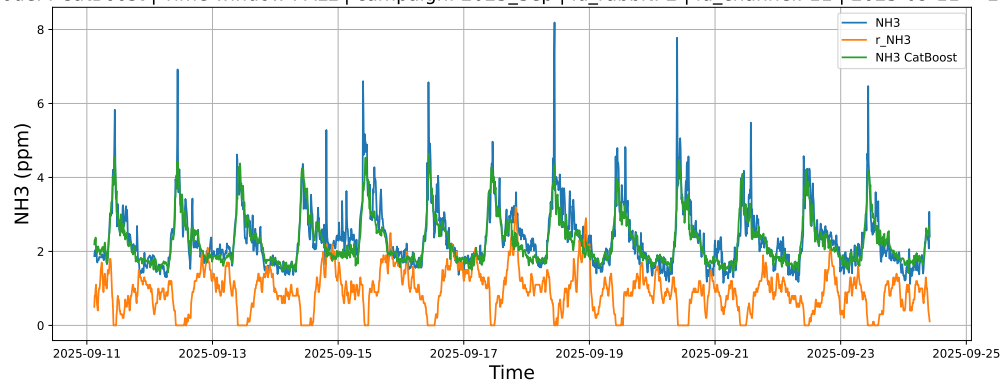
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



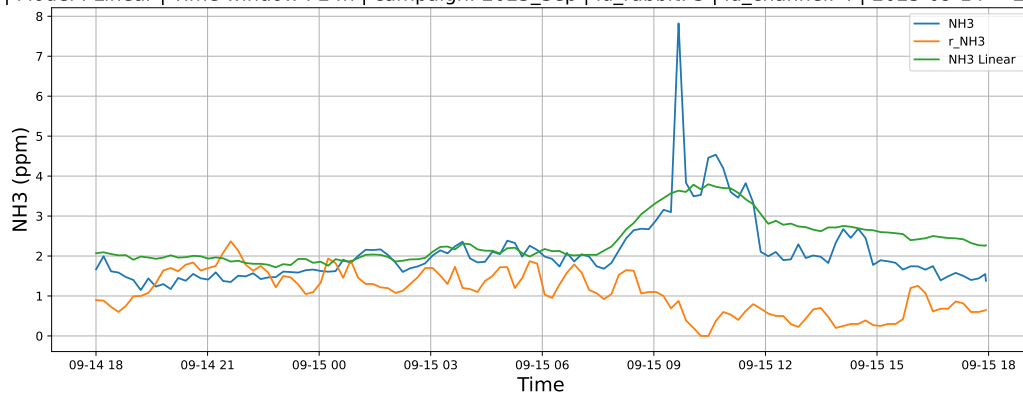
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



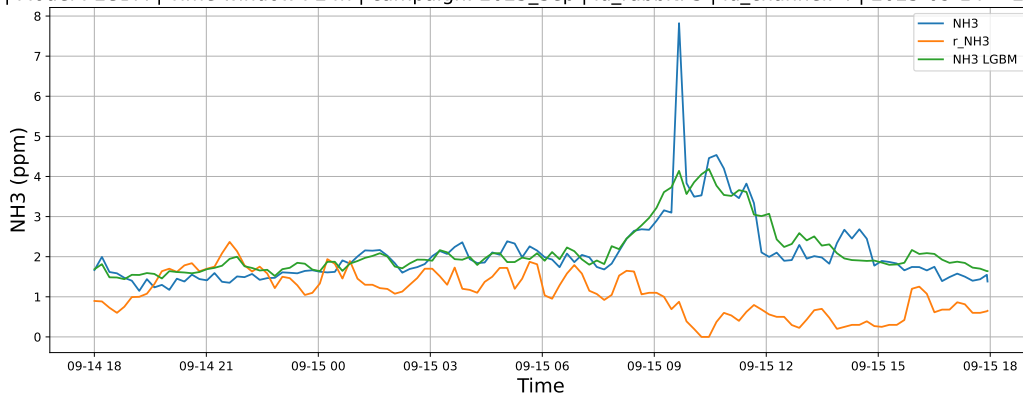
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

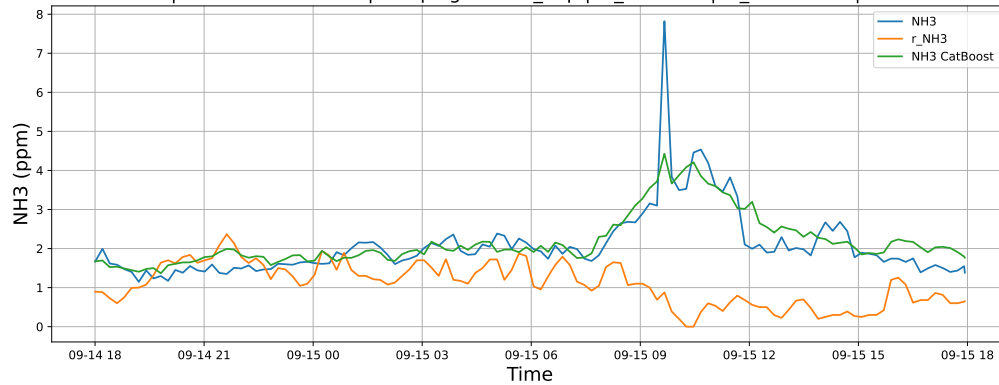
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

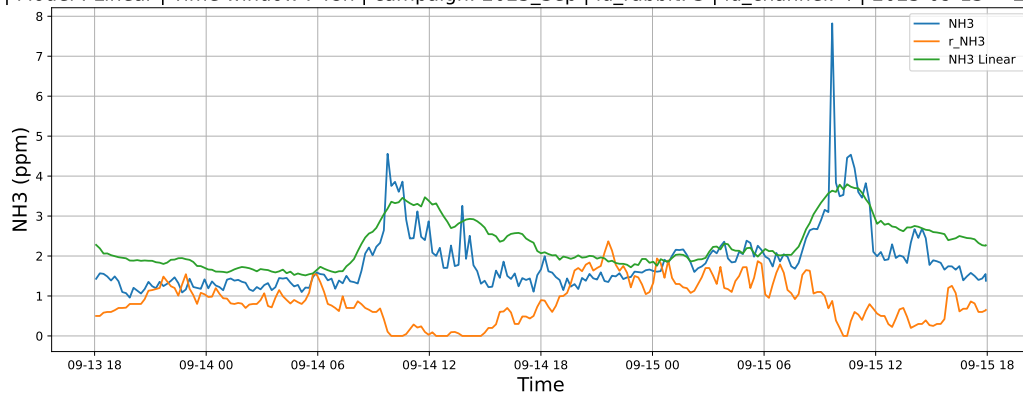


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

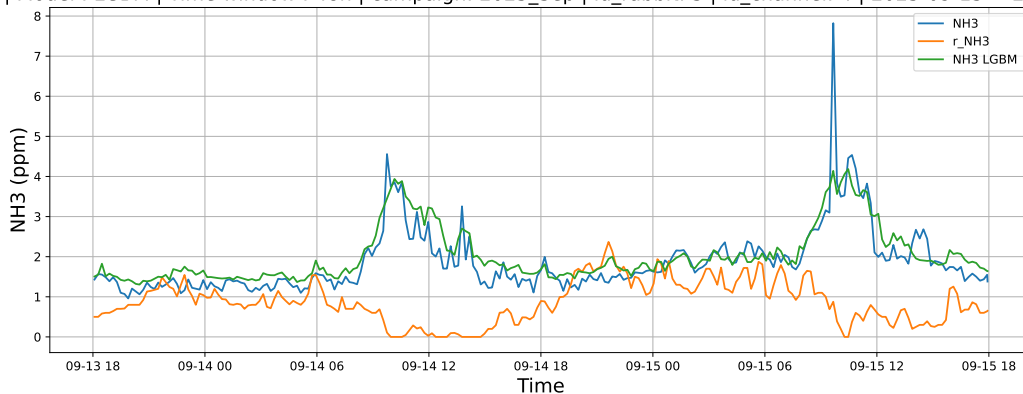


4.1.6.2 Time window : 48

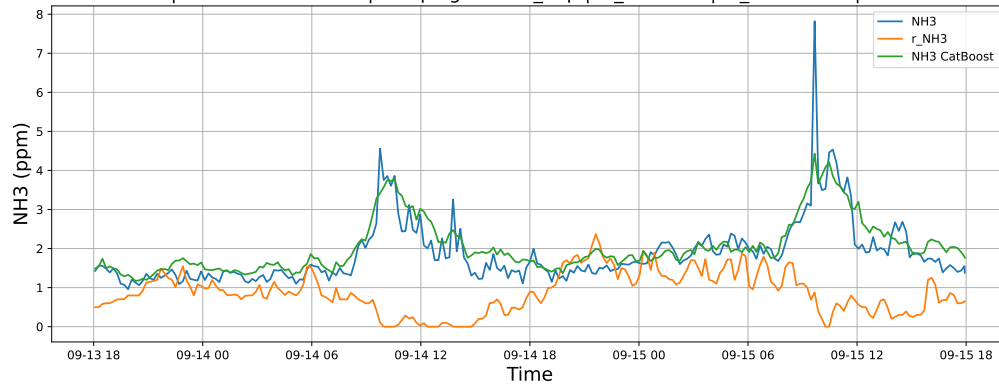
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

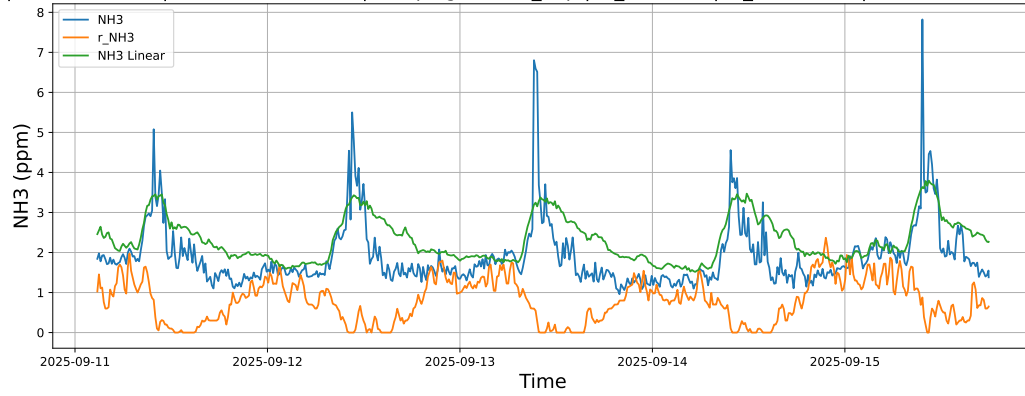


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

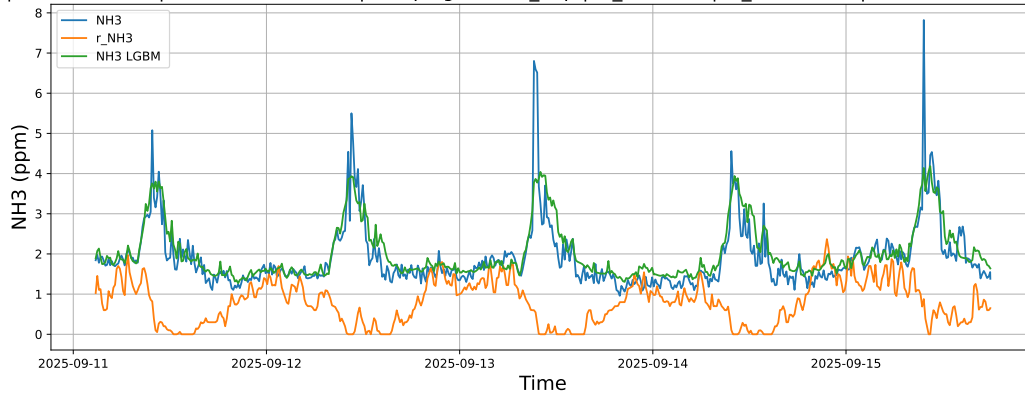


4.1.6.3 Time window : ALL

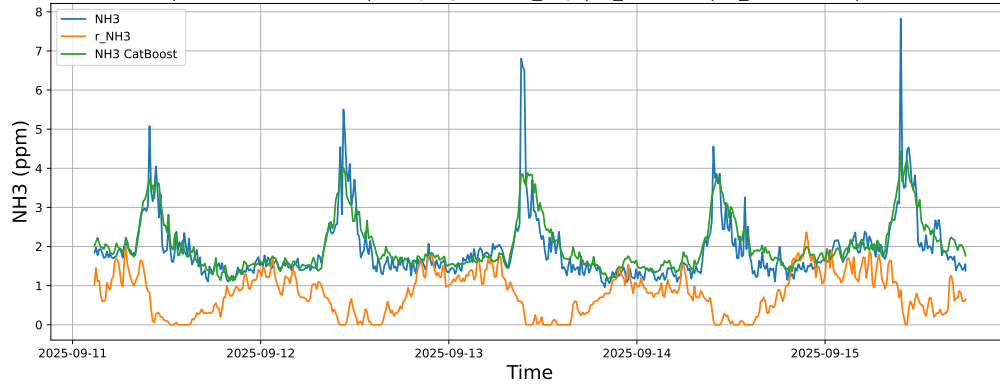
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



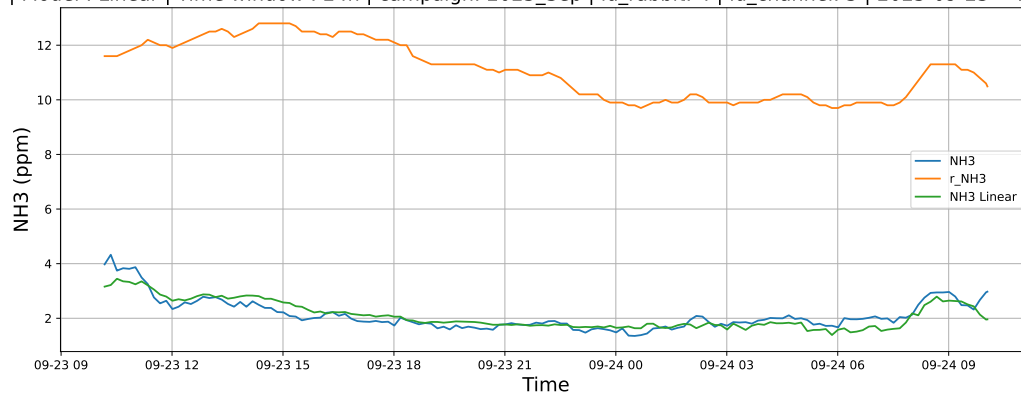
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



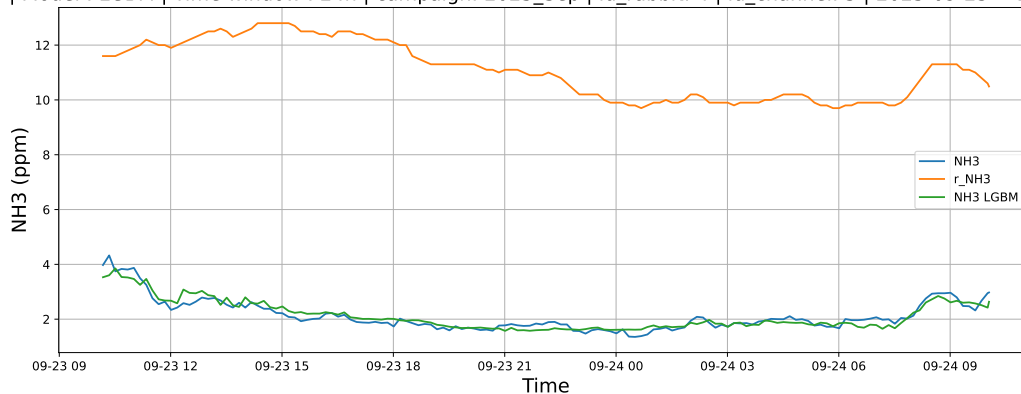
4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

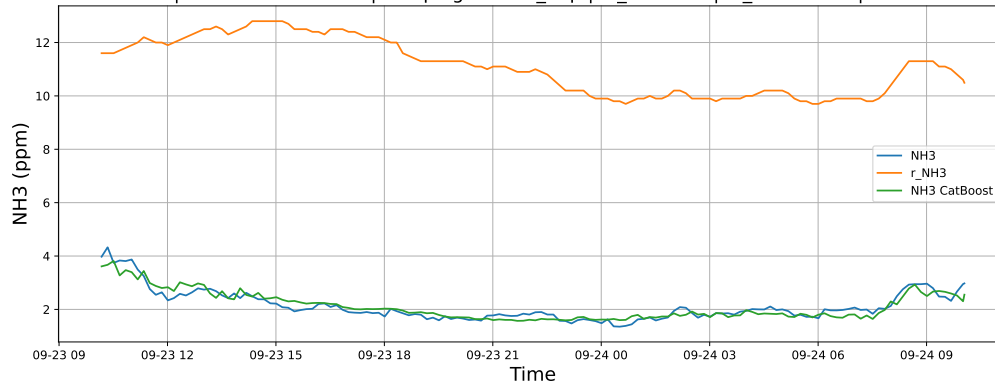
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

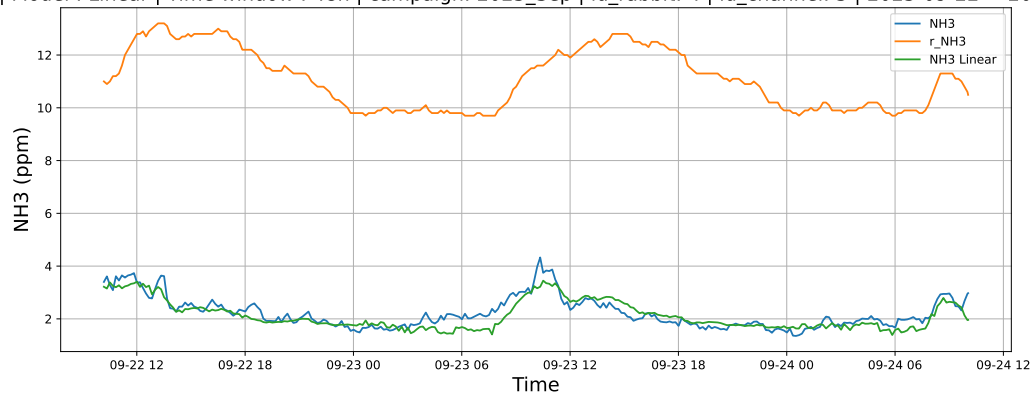


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

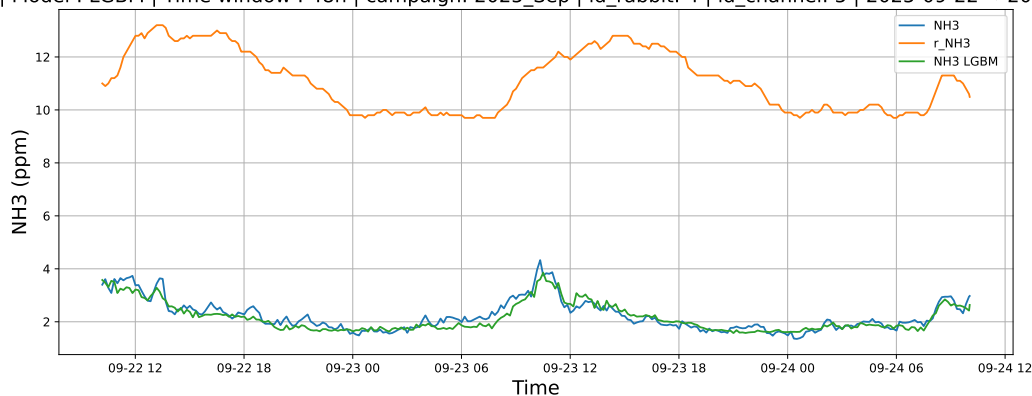


4.1.7.2 Time window : 48

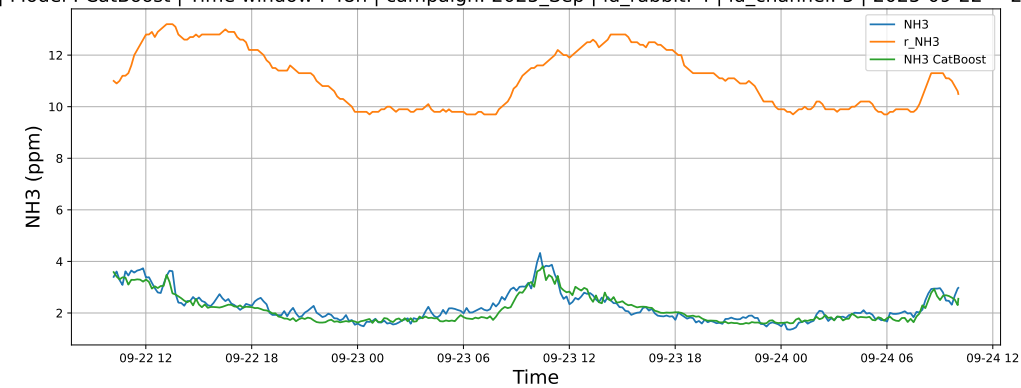
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

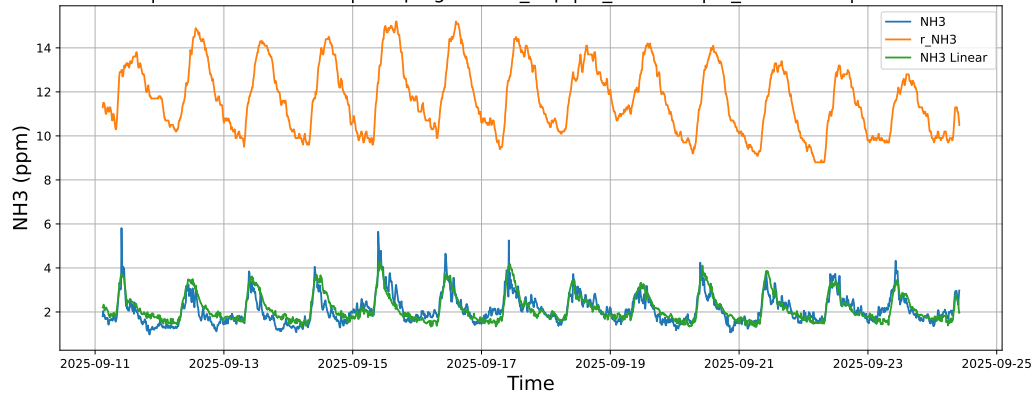


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

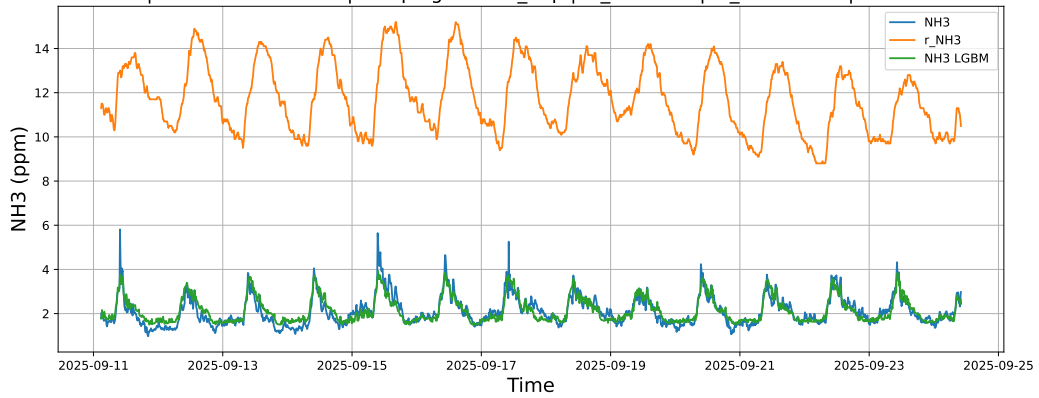


4.1.7.3 Time window : ALL

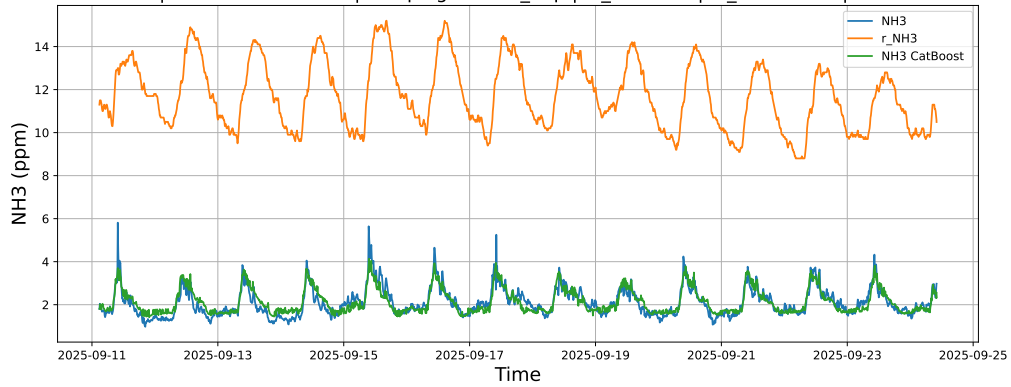
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



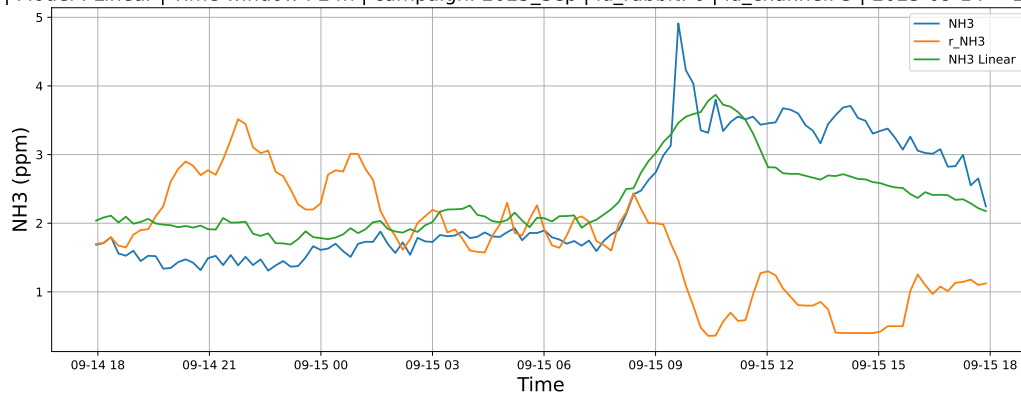
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



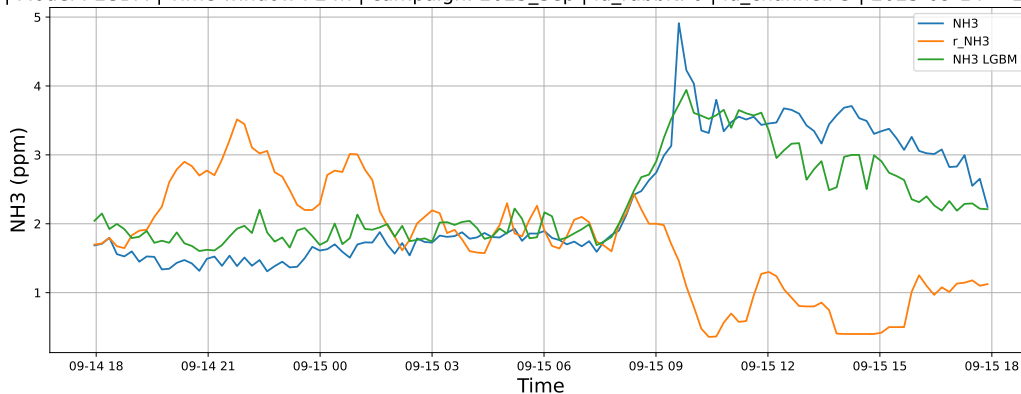
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

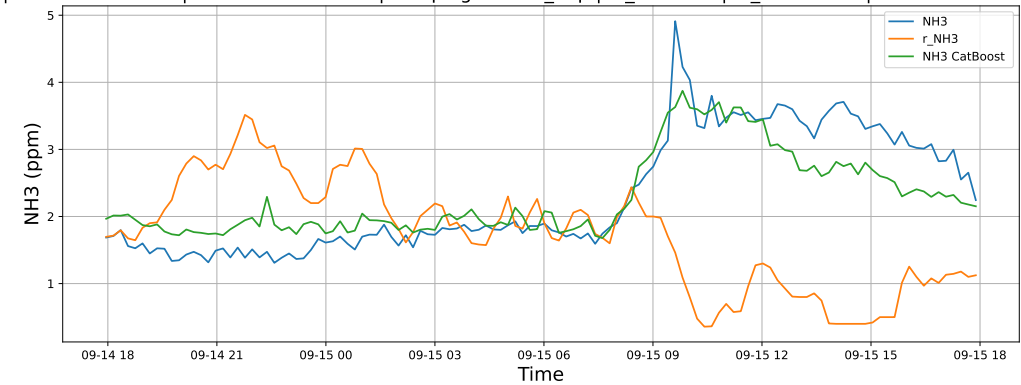
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

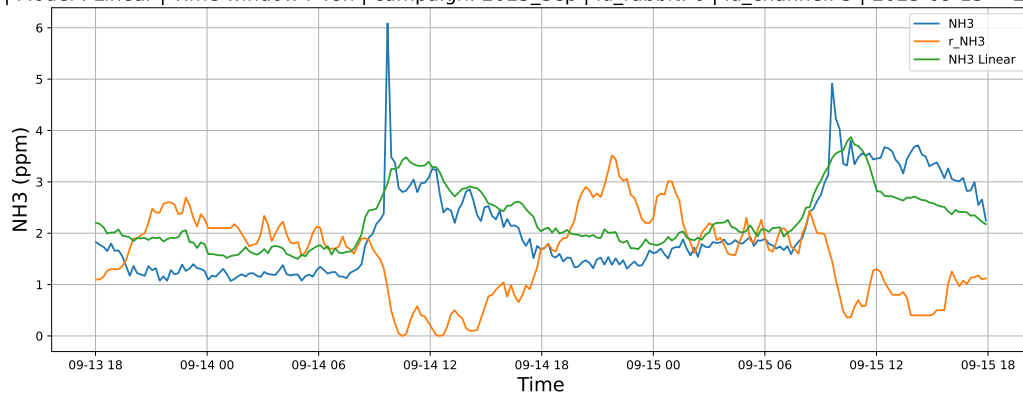


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

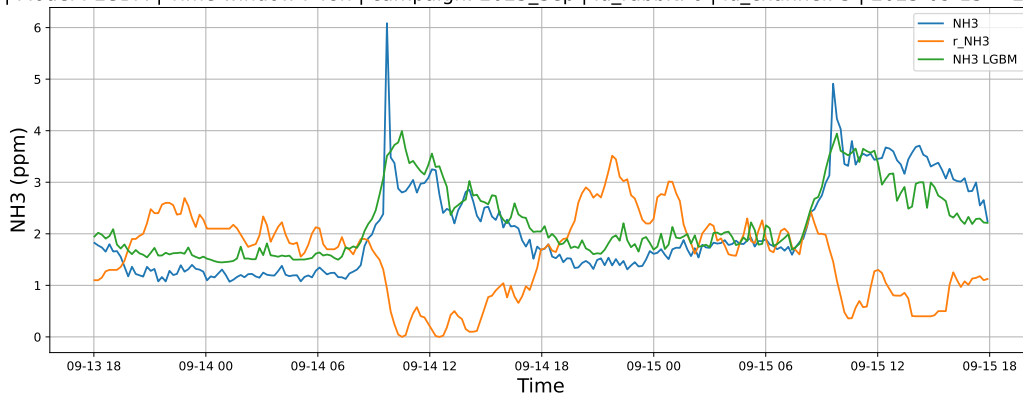


4.1.8.2 Time window : 48

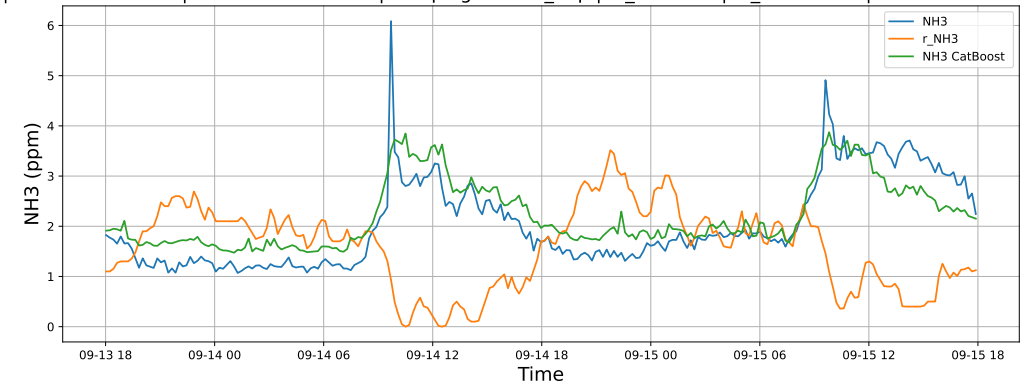
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

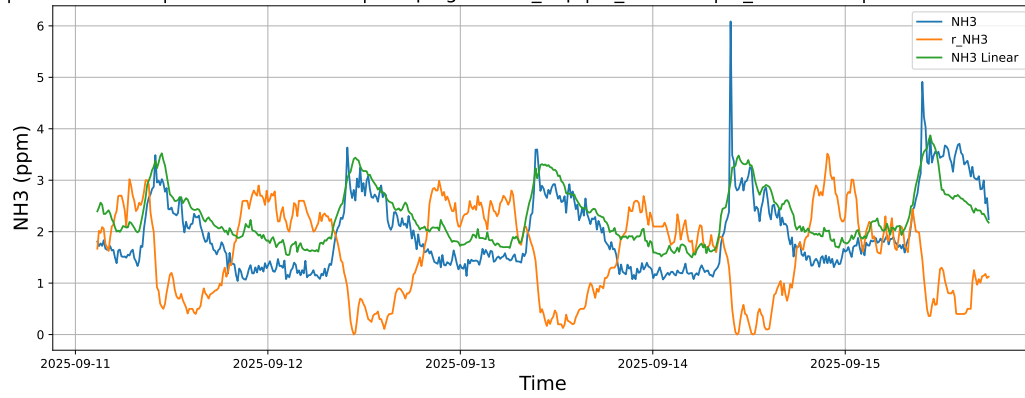


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

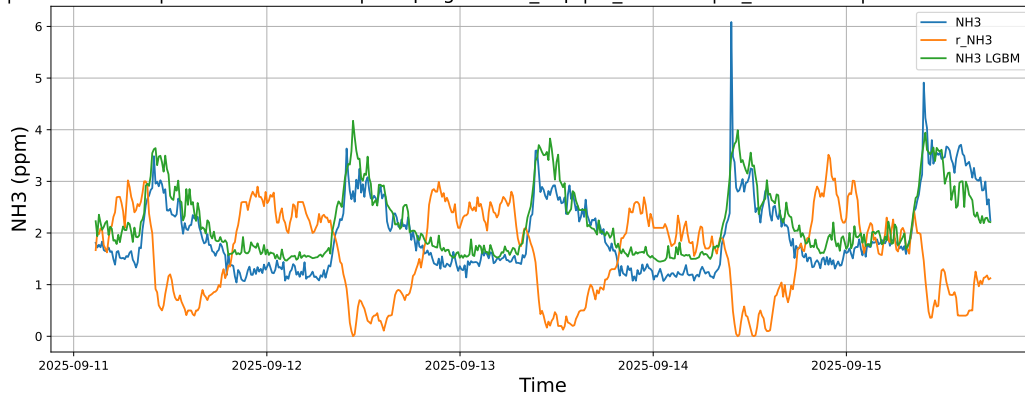


4.1.8.3 Time window : ALL

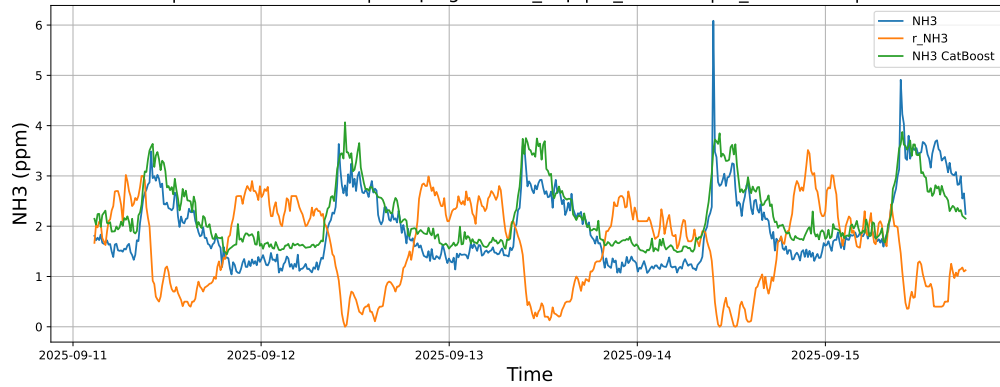
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

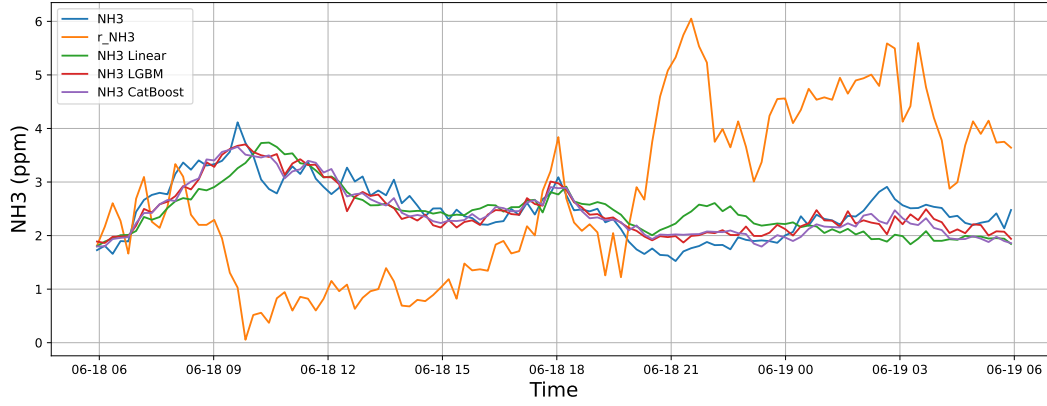


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

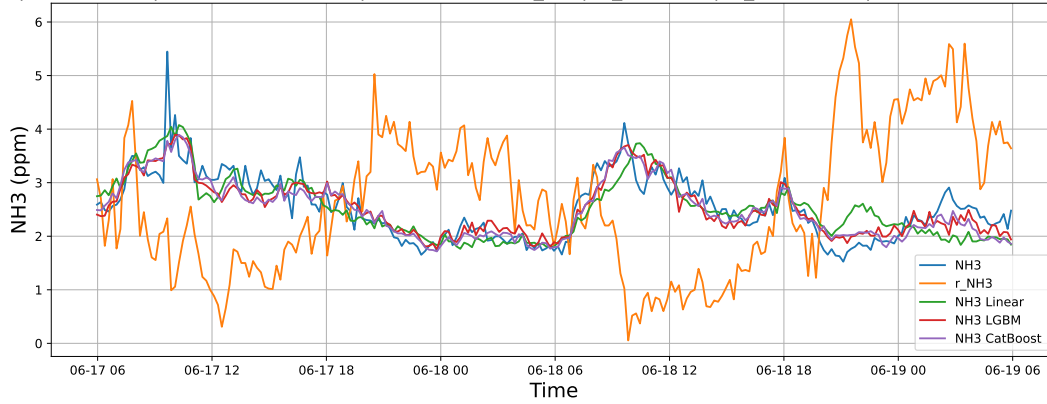
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



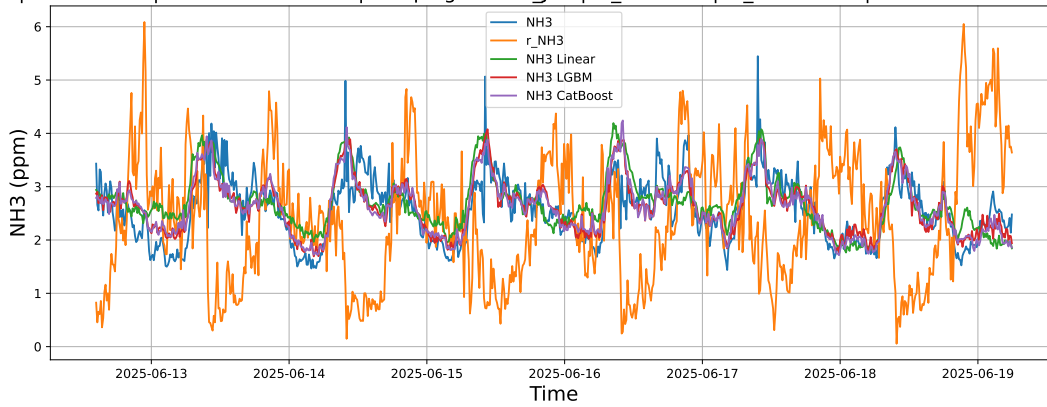
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

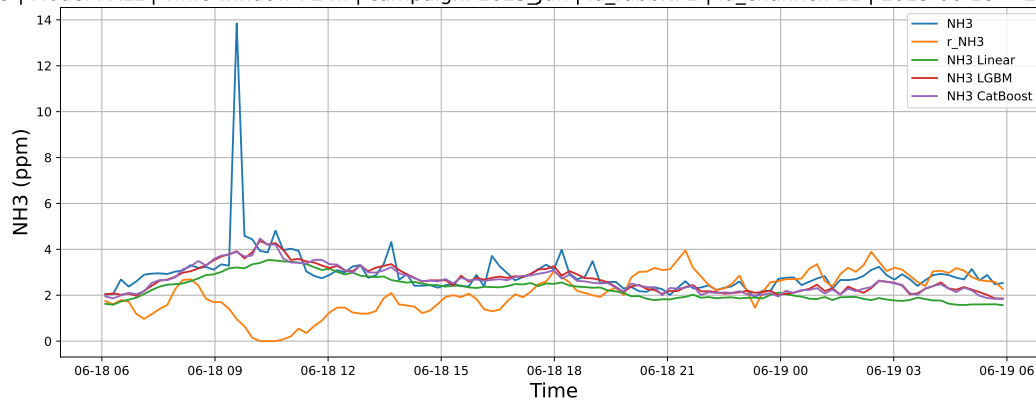
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

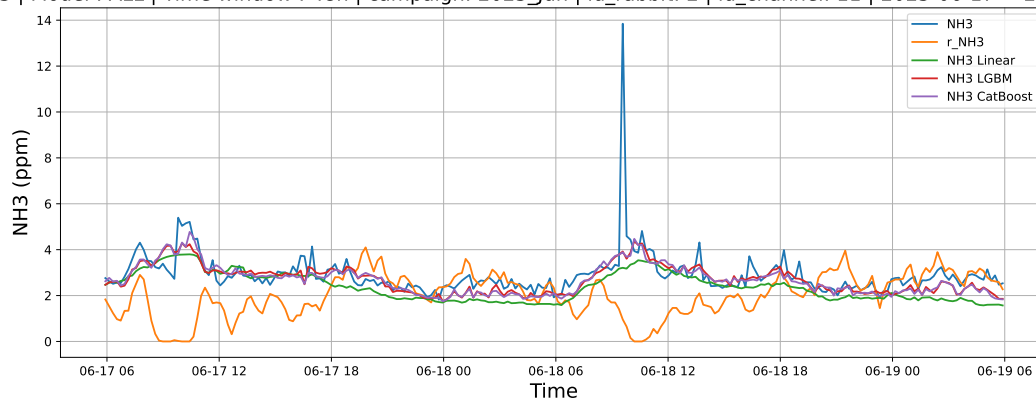
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



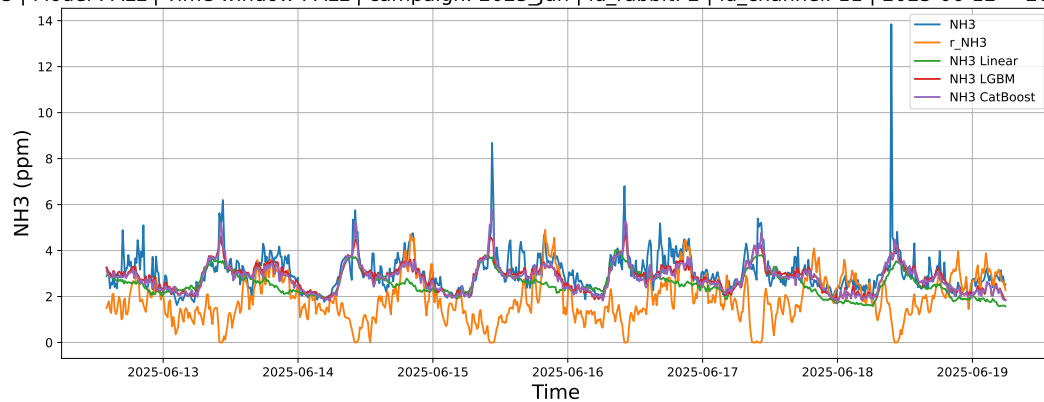
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

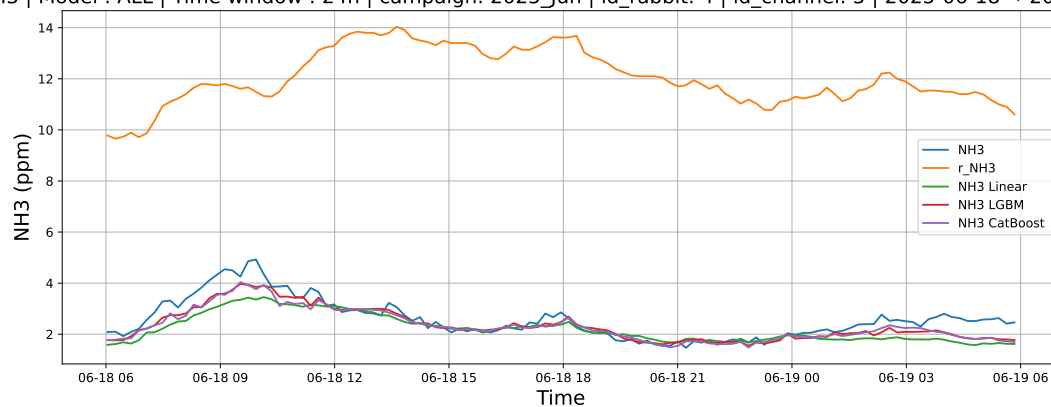
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

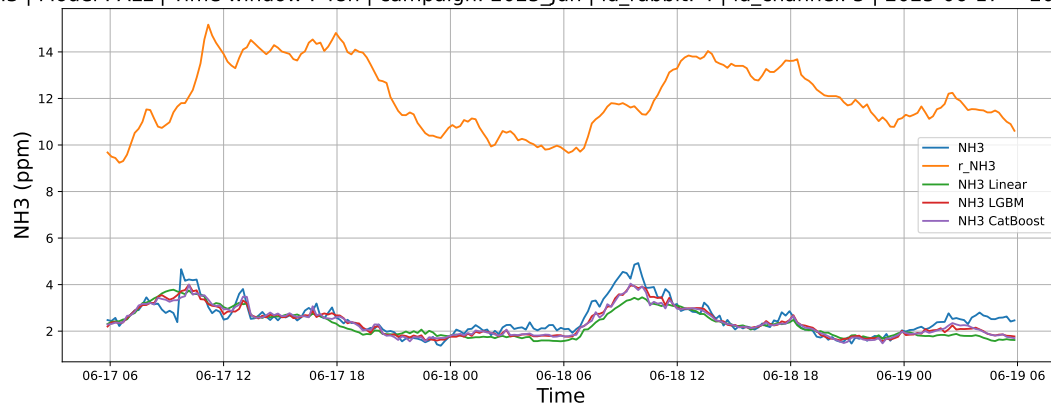
4.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



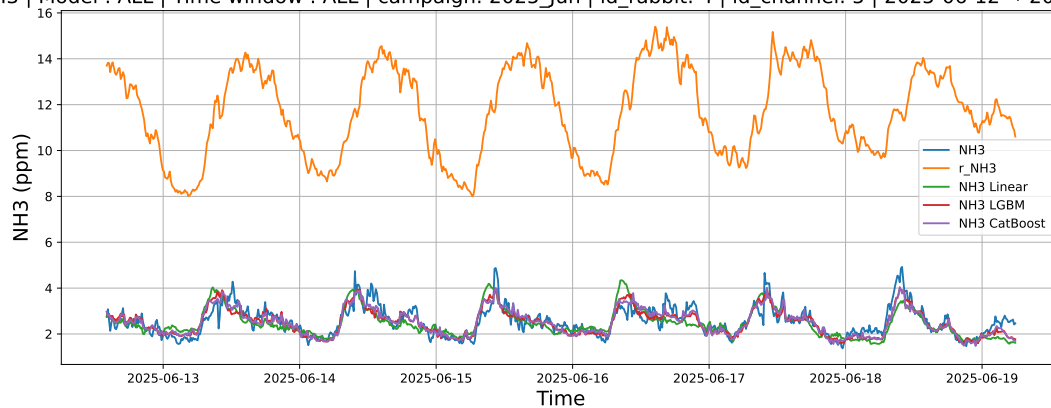
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.2.3.3 Time window : ALL

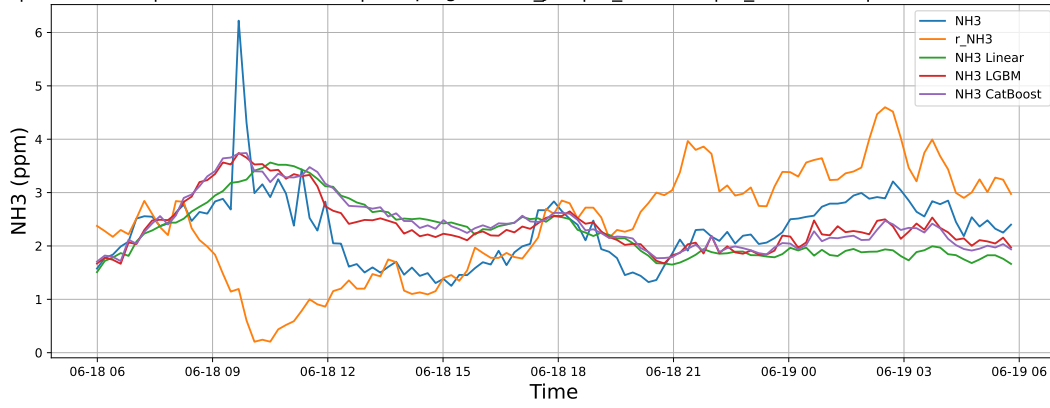
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

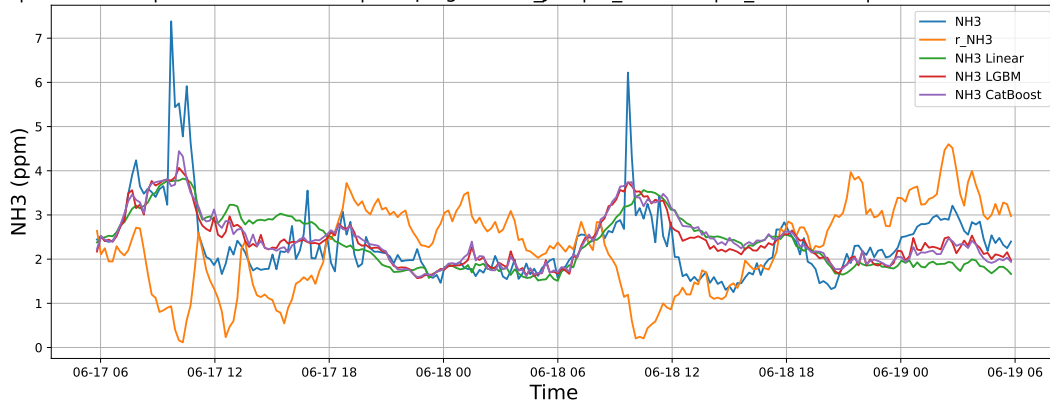
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



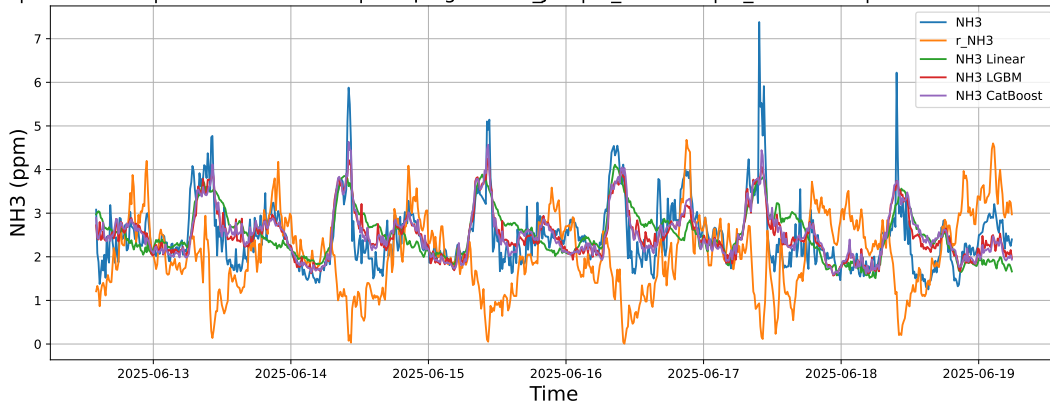
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.2.4.3 Time window : ALL

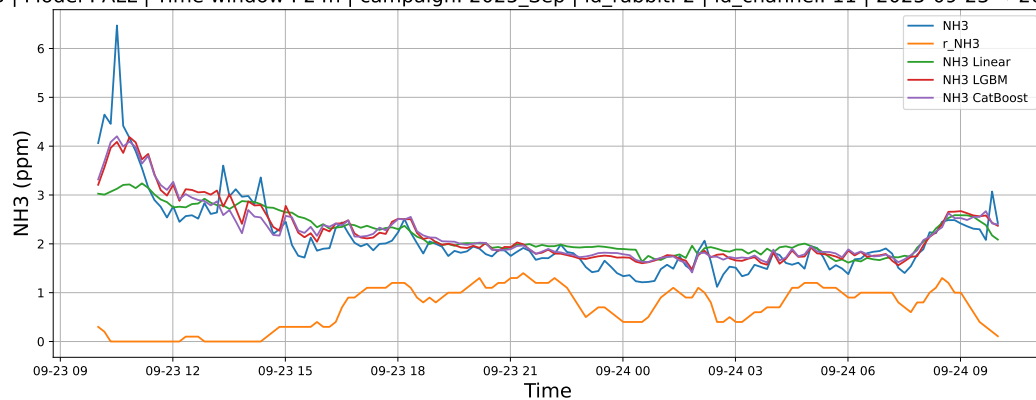
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

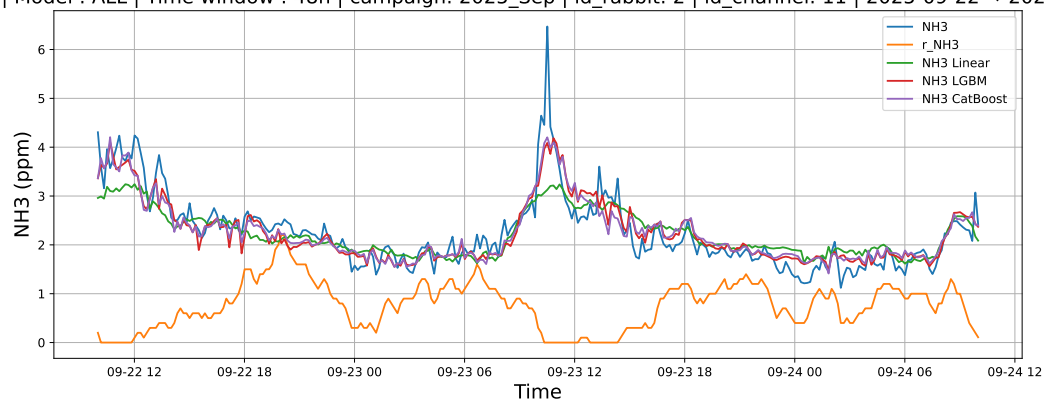
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



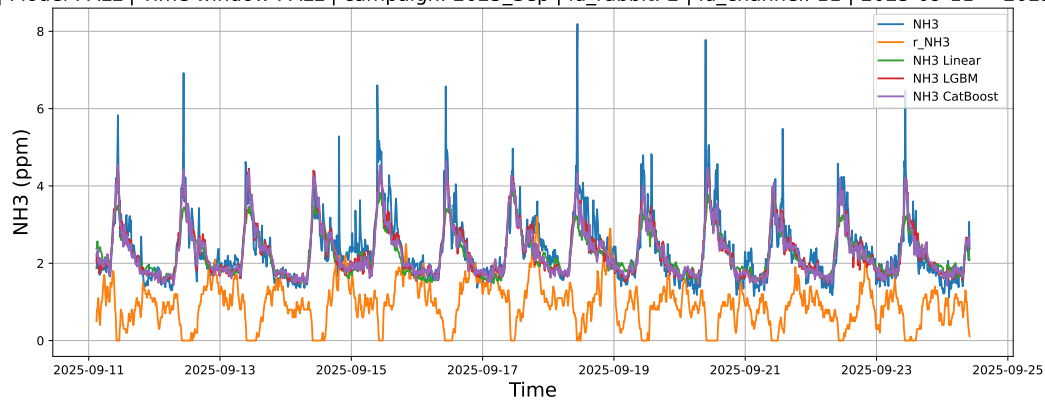
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

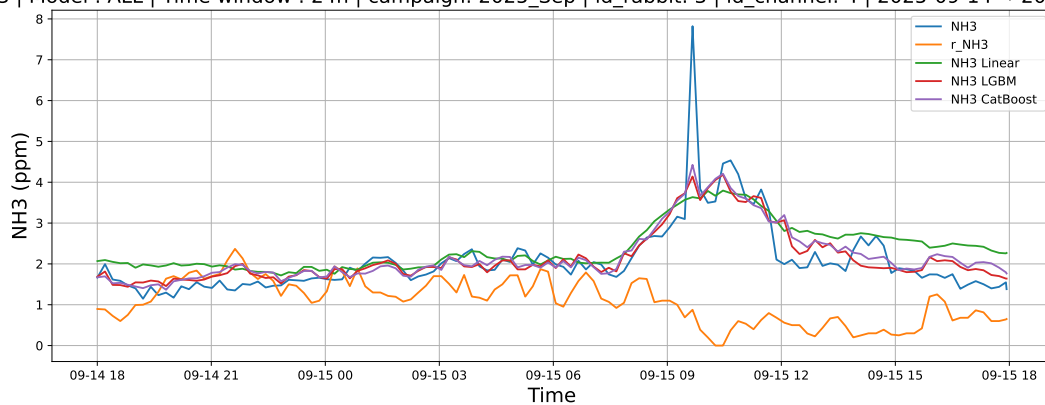
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

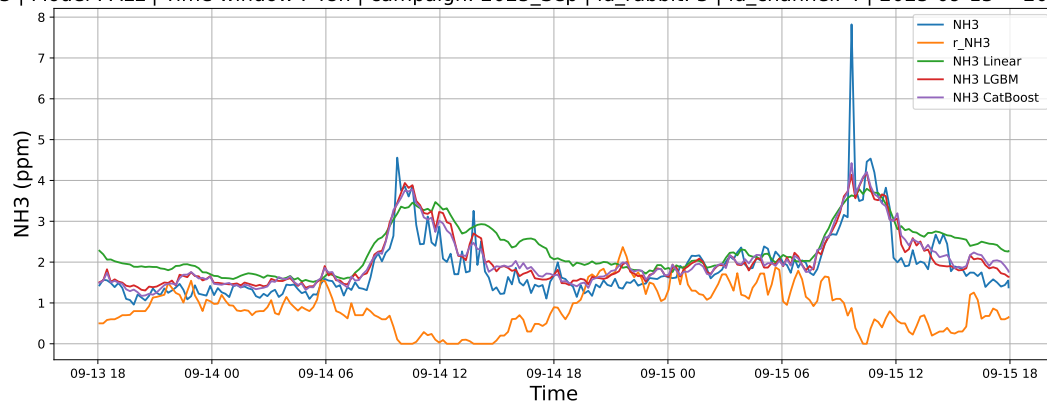
4.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



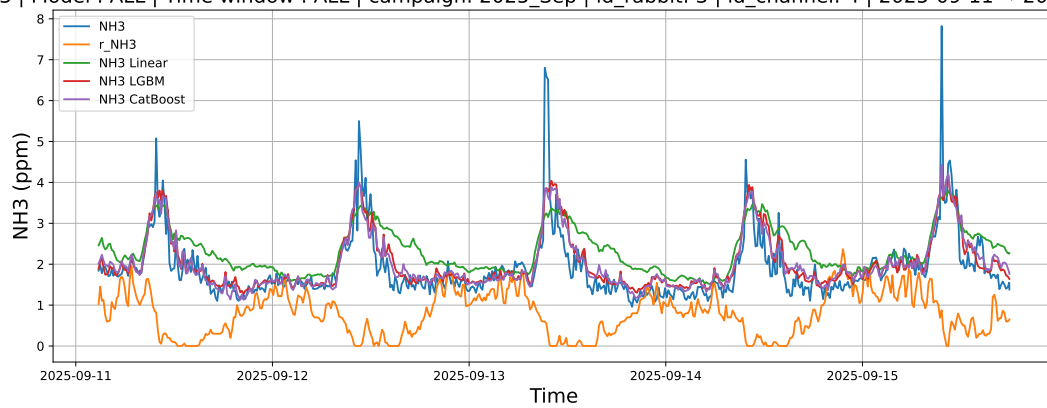
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.2.6.3 Time window : ALL

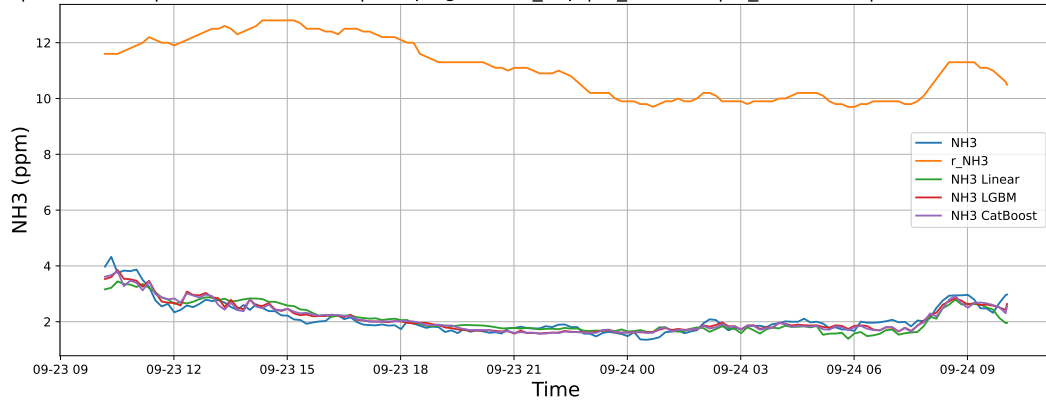
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

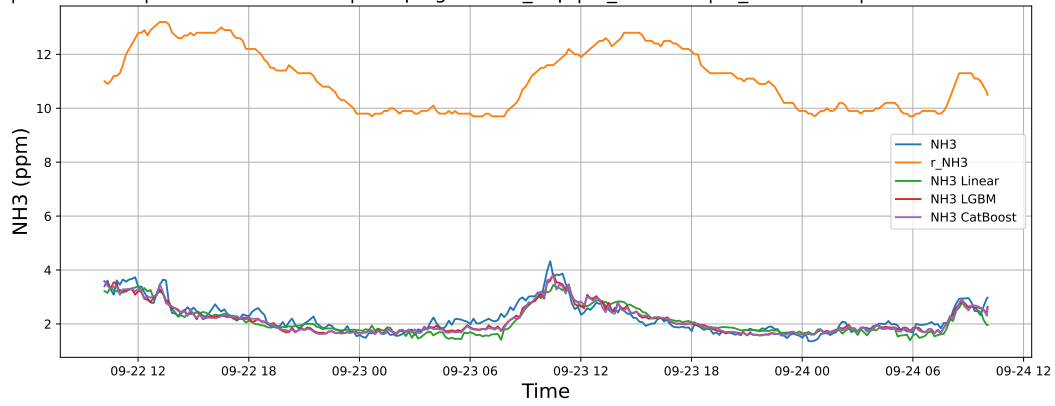
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



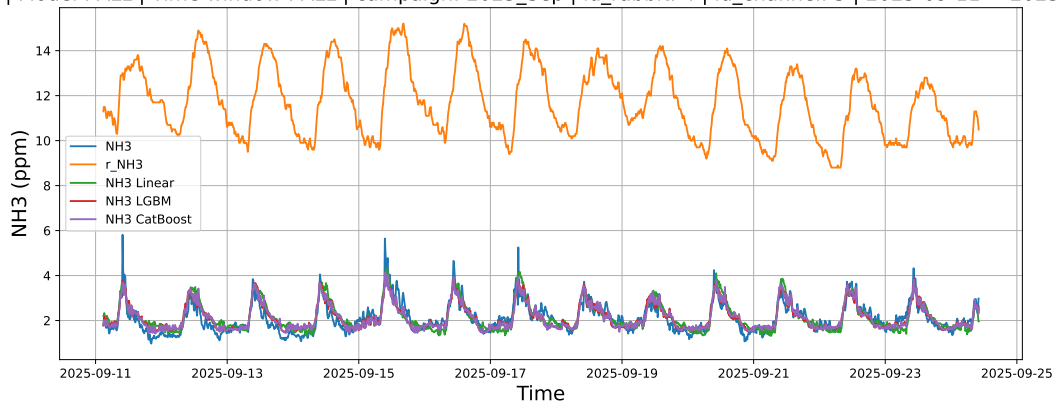
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



4.2.7.3 Time window : ALL

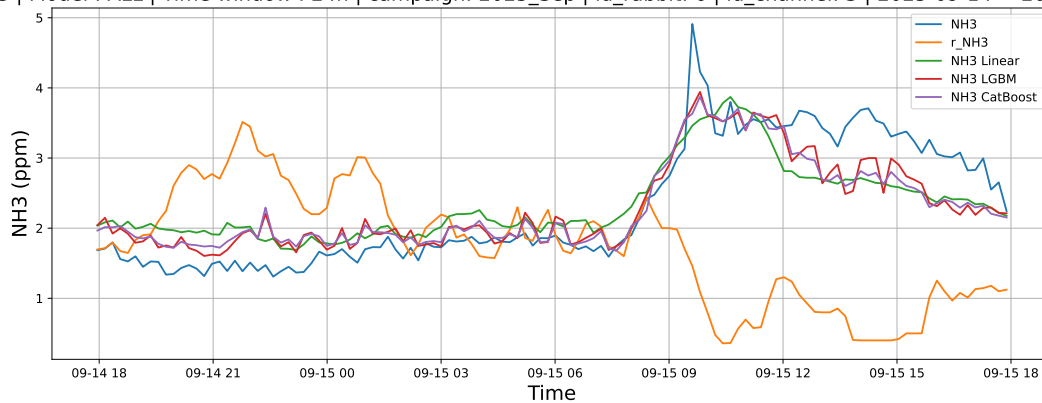
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

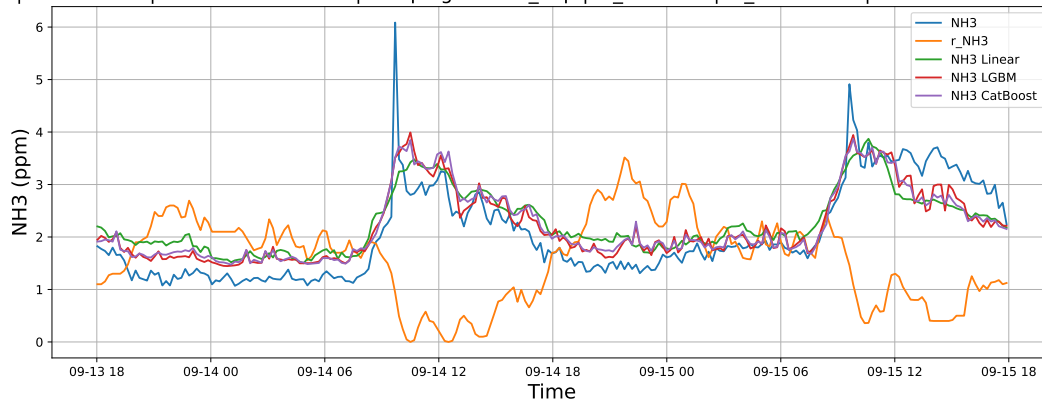
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

